What do mRNA-based predictions of microRNA expression measure? An attribution analysis of 337 miRNA precursors in TCGA breast tumours

Janaki Balasubramanian

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Keywords

MicroRNA · Breast cancer · TCGA · Host genes · Batch effects · Cross-validation · Tumour microenvironment

Abbreviations

CPE	Consensus purity estimate	PCA / PC	Principal component analysis / component
CV	Cross-validation	R^2	Coefficient of determination
ER	Oestrogen receptor	RF	Random forest
FFPE	Formalin-fixed, paraffin-embedded	RPM	Reads per million miRNA-mapped reads
GDC	Genomic Data Commons	SD	Standard deviation
IQR	Interquartile range	SHAP	SHapley Additive exPlanations
LUMP	Leukocytes unmethylation for purity	TCGA	The Cancer Genome Atlas
miRNA	MicroRNA	TCGA-BRCA	TCGA breast invasive carcinoma project
mRNA	Messenger RNA	TPM	Transcripts per million
PAM50	50-gene intrinsic subtype classifier	TSS	Tissue source site

Abstract

MicroRNA (miRNA) levels are increasingly estimated from mRNA expression when miRNA sequencing is not available, but it is rarely asked what such predictions capture. Matched mRNA and miRNA sequencing of 1,053 TCGA breast tumours was used to predict 337 well-expressed miRNA precursors from 27,671 genes with ridge regression. Each prediction was compared, on the same cross-validation folds, with simpler reference models built from the miRNA's host gene(s), the genes at its genomic locus, the leading 20 principal components of the transcriptome (interpreted as tumour state) and clinical covariates. The mRNA and miRNA libraries of each tumour had been sequenced on effectively the same plates (Cramér's V = 0.97). Holding out whole plates lowered the mean R^2 of the full model from 0.427 to 0.380 and the number of miRNAs with $R^2 > 0.5$ from 106 to 65, whereas holding out tissue source sites lowered it only to 0.413. Plate-grouped cross-validation was therefore used as the main design. Under it, 240 miRNAs reached a cross-validated R^2 of at least 0.3 with the best model. Among them, the 20 components reproduced a median of 67% of the best achievable R^2, the components with the host gene(s) 85%, and the components with the genes within ±500 kb 96%, whereas PAM50 subtype and purity reproduced 22%. The components correlated with ER/luminal, proliferation, basal, immune and stromal programmes; one of them mainly tracked sequencing plate, but removing plate effects from both data types did not reduce the share reproduced by the components. The number of miRNAs whose full prediction exceeded every reference model by at least 0.10 fell from 47 under patient-grouped folds to 25, and all 5 of these with a robust label are paralogous precursors. For 17 of 32 predictable paralogous precursors, genes at the paralog's locus predicted the precursor better than genes at its own locus, which cautions against locus-specific interpretation of precursor-level values. Labels of individual miRNAs were uncertain: 104 of 240 were robust to analysis settings and fold-to-fold variation. In bulk breast tumours, mRNA-based predictions of miRNA expression thus largely reproduce host-gene, locus and tumour-state information, and part of their apparent accuracy reflects processing batches shared by the two assays. Inferred miRNA levels should be evaluated against these baselines with batch-aware cross-validation before being interpreted as miRNA-specific biology.

Plain-language summary

Cells make thousands of messenger RNAs, which carry the instructions for proteins, and a smaller set of microRNAs, which switch some of those instructions down. Measuring microRNAs needs a separate experiment, so researchers sometimes estimate them from messenger RNA data instead. This project asked what such estimates really capture, using about a thousand breast tumours from a public cancer database.

Part of the apparent accuracy turned out to be technical. The tumours were processed in batches, and both kinds of RNA from the same tumour were processed in the same batch, so a model can learn a batch's fingerprint and use it to "predict" the microRNA. When whole batches were held back for testing, accuracy dropped noticeably. Of what remained, most could be reproduced by much simpler estimates. Many microRNAs are made from inside another gene, so when that gene is active the microRNA is made too, and that gene, together with the overall make-up of the tumour (how many immune cells it contains, whether it is hormone-sensitive, how fast it is growing), was usually enough. Only a few microRNAs carried information that these simpler estimates did not capture, and most of those exist as near-identical copies in the genome, where the measured value of one copy was often better predicted by genes near the other copy. These results help researchers judge when an estimated microRNA level says something about the microRNA itself and when it may be a stand-in for something simpler.

Introduction

MicroRNAs are small non-coding RNAs, about 22 nucleotides long, that guide Argonaute proteins to partially complementary sites in target mRNAs and reduce their translation and stability (Bartel, 2018). They are transcribed as longer primary transcripts, and most miRNA genes lie within other transcription units: in introns or exons of protein-coding genes, in dedicated non-coding host transcripts, or in clusters that are transcribed together (Rodriguez et al., 2004). Intronic miRNAs are frequently co-expressed with their host genes (Baskerville & Bartel, 2005), although a pan-cancer analysis of TCGA found that only a minority of miRNA–host pairs are consistently co-expressed across cancer types (Liu et al., 2021). A miRNA's level is therefore tied to the transcription of other genes for reasons unrelated to its regulatory function.

Matched mRNA and miRNA profiles are available for large tumour cohorts, including more than a thousand breast tumours in TCGA (Cancer Genome Atlas Network, 2012; Grossman et al., 2016). This has encouraged models that predict miRNA expression from gene expression. A pan-cancer study predicted 1,300 miRNAs from TCGA gene expression and found that ridge regression reached test $R^2 > 0.5$ for 353 of them, outperforming lasso and random forests (Cihan et al., 2025). Tools such as miRSCAPE and SiCmiR train on TCGA and infer miRNA expression from bulk or single-cell RNA-seq, so that miRNA activity can be studied where it was never measured (Olgun et al., 2022; Cai et al., 2025). Such predictions are useful only if it is clear what they capture.

Bulk tumour samples mix cancer cells with immune, stromal and endothelial cells, and breast tumours differ systematically by intrinsic subtype (Perou et al., 2000; Parker et al., 2009) and proliferation. Many miRNAs are cell-type-specific, for example miR-150 in mature lymphocytes (Xiao et al., 2007), so a model can predict a miRNA by measuring the cell composition of the sample rather than anything specific to the miRNA. Similarly, a model can predict an intronic miRNA by reading its host transcript. A further possibility is technical. Large consortium datasets are generated in batches, and batch effects are widespread in high-throughput data (Leek et al., 2010). When batch structure is shared between the features and the outcome and the cross-validation does not hold batches out, performance estimates can be strongly biased (Soneson et al., 2014). Whether the mRNA and miRNA assays in TCGA share processing batches, and whether this affects the accuracy of mRNA-based miRNA prediction, has not, to my knowledge, been examined. High predictive accuracy is therefore compatible with several explanations: host-gene or local co-transcription, tumour subtype and cell composition, shared technical batches, or a genuine miRNA-specific relationship.

This project asks what mRNA-based predictions of miRNA expression measure in breast tumours. Instead of interpreting a single model, it compares the prediction of each miRNA with a set of simpler reference models, each representing one candidate proxy, and asks which simplest model is sufficient to reproduce most of the prediction. The objectives were to: (1) test whether shared sequencing batches inflate the cross-validated accuracy of mRNA-based miRNA prediction; (2) measure how well each well-expressed miRNA precursor can be predicted when whole batches are held out; (3) quantify how much of that performance is reproduced by the host gene, the local genomic neighbourhood, the overall tumour state and clinical covariates; (4) identify the biological programmes behind tumour-state miRNAs; (5) examine paralogous precursors, whose quantification may not be locus-specific; and (6) assess how robust the attribution of individual miRNAs is.

Results

Study design and cohort

The analysis workflow is shown in Figure 1. Of the 1,087 TCGA-BRCA primary tumours with both mRNA and miRNA sequencing, 13 formalin-fixed samples were removed, leaving one tumour per patient (1,074). Tumours with a PAM50 call and a consensus purity estimate were retained, giving an analysis set of 1,053 tumours (Table 1). Prediction targets were the 337 miRNA precursors with RPM > 1 in at least half of the tumours, and features were the 27,671 genes with TPM > 0.5 in at least 10% of tumours. Every model was evaluated on the same 15 test folds, in which whole sequencing plates were held out (5 folds × 3 repeats; the reason is given in the next section), so that their R^2 values can be compared directly for each miRNA. For each miRNA, the ceiling is the highest cross-validated R^2 of the full model or any reference model, and attribution means that a specified reference model reproduces at least 80% of that ceiling. The categories are hierarchical and overlap in the information they use; they describe predictive sufficiency, not causal contributions to expression.

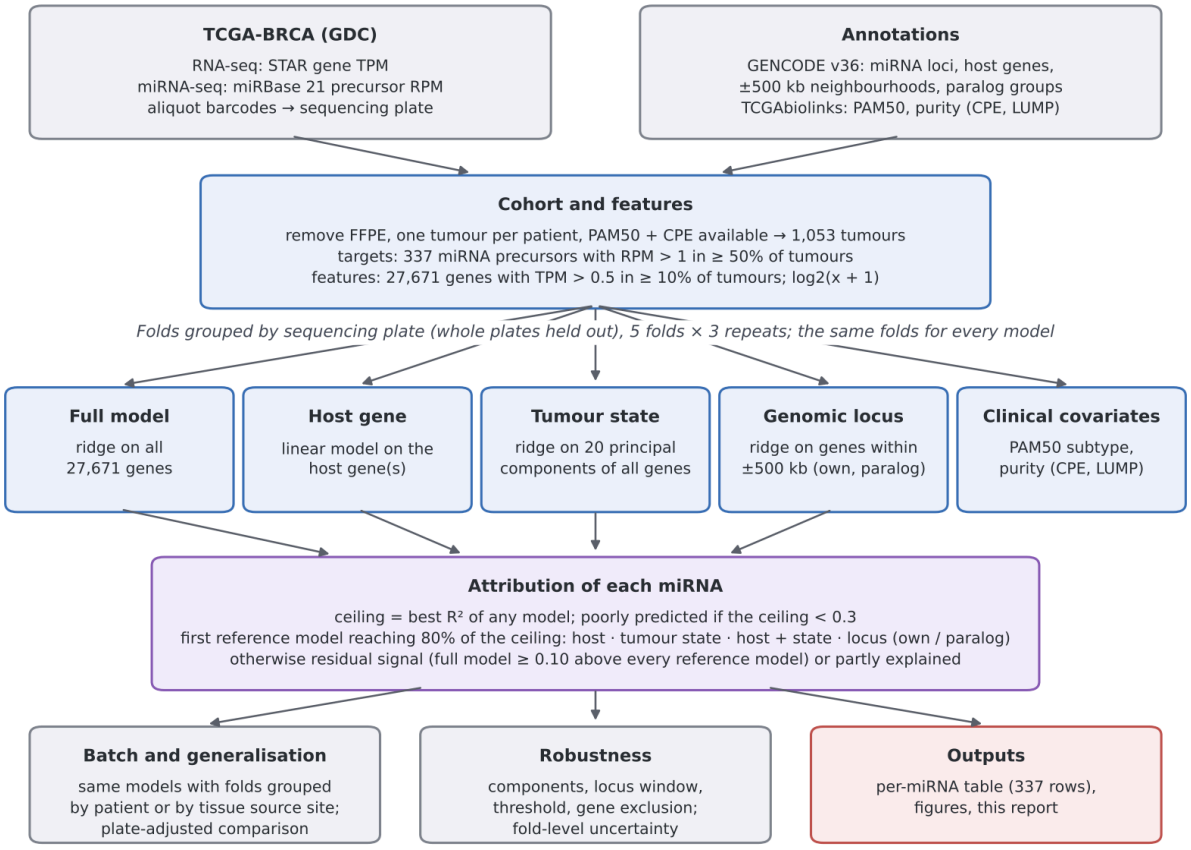


Figure 1. Analysis workflow. A full ridge model on all genes is compared, for each miRNA and on identical plate-grouped cross-validation folds, with reference models representing the host gene, the genomic locus, low-dimensional transcriptomic structure (interpreted as tumour state) and clinical covariates. Each miRNA is then assigned to the first reference model, in a fixed order, that reproduces at least 80% of the best R^2 reached by any model.

Table 1. Cohort, targets, features and cross-validation design.

Item	Value
Primary tumours with mRNA and miRNA sequencing	1,087
After removing FFPE samples (one tumour per patient)	1,074
With a PAM50 call	1,062
With a CPE purity estimate (analysis set)	1,053
PAM50 subtypes in the analysis set	LumA 551, LumB 200, Basal 181, Her2 81, Normal 40
Sequencing plates (mRNA / miRNA); tumours per plate	40 / 40; median 17, range 1–89
Tissue source sites	40
miRNA precursors with RPM > 1 in ≥ 50% of tumours (targets)	337
… with a host gene: measurable / not measurable; no host gene	224 / 45; 68
… in a paralog group	61
Genes with TPM > 0.5 in ≥ 10% of tumours (features)	27,671
Cross-validation (main)	5 folds grouped by sequencing plate, 3 repeats (15 test folds of 210–211 tumours)

Shared sequencing batches inflate patient-grouped cross-validation

The mRNA and miRNA libraries of TCGA tumours were sequenced in batches, identified by the plate field of the aliquot barcode (40 plates per assay; median 17 tumours per plate in the analysis set, range 1–89). The plates of the two assays were almost interchangeable: of the 1,600 possible combinations of mRNA and miRNA plate, only 47 occurred (Cramér's V = 0.97), so any plate effect applies to both measurements of the same tumour. Plate was only partly related to tissue source site (V = 0.38; 0.34 after bias correction). Under patient-grouped cross-validation, a common design, tumours from the same plate appear in both the training and the test folds. To test whether this inflates accuracy, the full model and the 20-component model were refitted with folds that held out whole plates or whole tissue source sites, each balanced by tumour count (Table 2 and Figure 2).

Holding out plates lowered the mean pooled out-of-fold R^2 of the full model from 0.427 to 0.380, a loss of 11%, and reduced the number of miRNAs with $R^2 > 0.5$ from 106 to 65; 30 miRNAs lost more than 0.1. Holding out tissue source sites cost much less (0.427 to 0.413; 1 miRNA lost more than 0.1). The 20-component model lost less than the full model under plate-grouped folds (0.279 to 0.259). Among the largest losses were miRNAs that, under patient-grouped folds, seemed to carry signal beyond every simpler model: the mean per-fold R^2 of mir-143 fell from 0.37 to 0.01, of mir-326 from 0.32 to 0.10 and of mir-330 from 0.46 to 0.29. Part of the accuracy measured with patient-grouped folds therefore comes from batch structure shared by the two assays, and the full model, with its 27,671 genes, exploits it more than the components do. All further results use plate-grouped folds; patient-grouped values are given for comparison in the tables and in the results file.

Table 2. Mean R^2 over the 337 miRNAs under three cross-validation designs (5 folds, 3 repeats each; site and plate folds balanced by tumour count). Per fold, the mean of the 15 test-fold R^2 values; pooled, R^2 of all out-of-fold predictions of a repeat, averaged over repeats.

Model	Folds grouped by	Mean R^2, per fold	Mean R^2, pooled	miRNAs with pooled $R^2 > 0.5$
Full model (27,671 genes)	patient	0.423	0.427	106
	site	0.402	0.413	96
	plate	0.362	0.380	65
Tumour state (20 components)	patient	0.275	0.279	16
	site	0.257	0.270	16
	plate	0.239	0.259	16

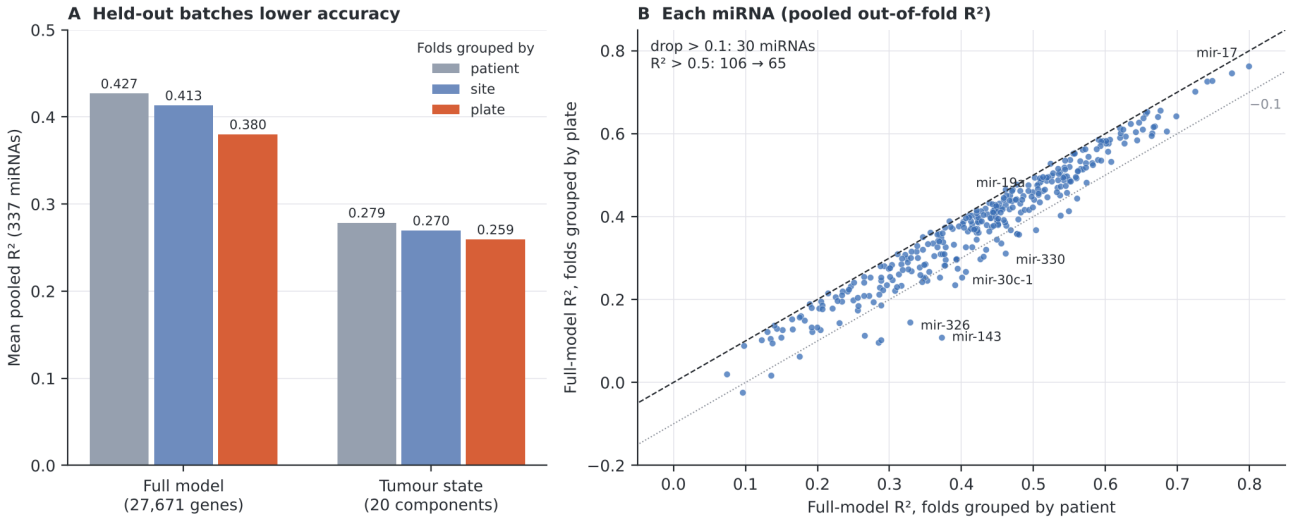


Figure 2. Effect of holding out batches. (A) Mean pooled out-of-fold R^2 of the full model and the 20-component model with folds grouped by patient, tissue source site or sequencing plate. (B) Full-model R^2 of each miRNA with patient-grouped versus plate-grouped folds; dashed line, equal R^2; dotted line, a loss of 0.1.

Most well-expressed miRNA precursors remain predictable across batches

With plate-grouped folds, ridge regression on all genes predicted the 337 miRNAs with a mean cross-validated R^2 of 0.36 (median 0.37); 225 reached $R^2 \geq 0.3$ and 62 exceeded 0.5 (compared with 102 under patient-grouped folds). The best-predicted miRNAs included mir-17 (0.75), mir-150 (0.74), mir-190b (0.72), mir-18a (0.72) and mir-155 (0.70). For 140 miRNAs a reference model predicted at least as well as the full model, and for 127 it was better by more than 0.01. The full model is therefore not a ceiling, possibly because a single strong predictor such as a host gene is diluted when the penalty is spread over 27,671 genes. The best R^2 of any model, the ceiling, was at least 0.3 for 240 miRNAs (Figure 3), including 15 whose full-model R^2 was below 0.3. The remaining 97 miRNAs are labelled poorly predicted and are not attributed further. All 54 miRNAs that were poorly predicted under patient-grouped folds remained so, and 43 more joined them.

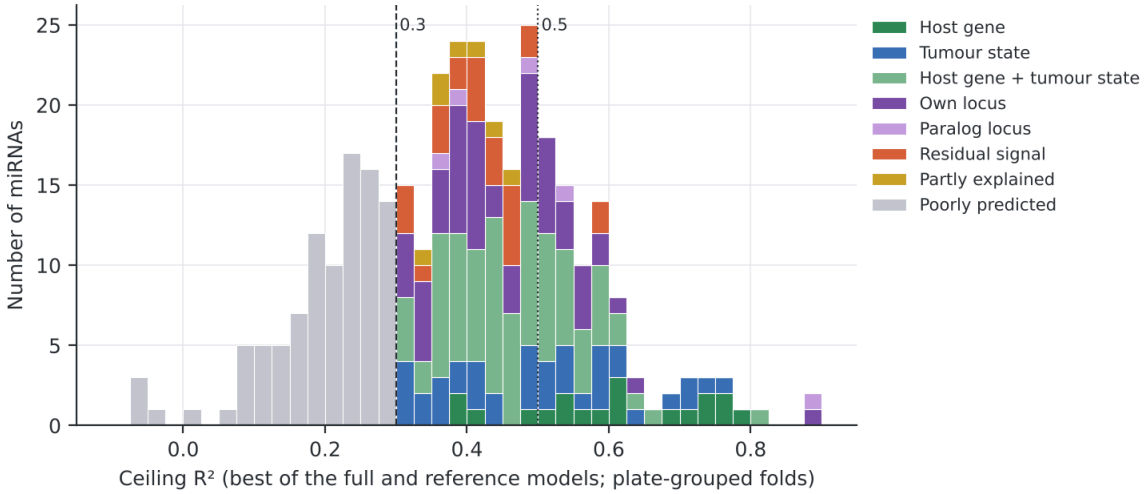


Figure 3. Ceiling R^2 (the best cross-validated R^2 of the full model or any reference model) for the 337 miRNA precursors, coloured by attribution category (defined in Table 4). Dashed line, $R^2 = 0.3$ (threshold for attribution); dotted line, $R^2 = 0.5$.

Simple reference models reproduce most of the transferable performance

To measure how much of each miRNA's predictive performance can be reproduced by the dominant structure of the transcriptome, the gene matrix was reduced to its leading principal components (computed within each training fold) and ridge regression was fitted on these components alone. For the 240 predictable miRNAs, 20 components reproduced a median of 67% of the ceiling, rising from 36% with 5 components to 77% with 50 (Figure 4A). Adding the host gene(s) raised the median to 85%, and adding the genes within ±500 kb of the miRNA to 96% (Figure 4B). These are fractions of predictive performance, not fractions of miRNA expression or of its regulation. The host gene(s) alone reproduced a median of 26% for the 167 predictable miRNAs with a measurable host gene, with a wide spread (Figure 4B). Clinical covariates reproduced much less: PAM50 subtype and the CPE purity estimate a median of 22% (at most 76% for any miRNA), and adding the LUMP estimate with a missingness indicator 22%. Subtype labels and purity estimates therefore capture only part of the structure that the transcriptome provides.

With patient-grouped folds, the corresponding medians were 64% for the components, 79% with the host gene(s) and 89% with the locus genes. The shares are higher with plate-grouped folds because the ceiling is lower: the accuracy that is lost when plates are held out is lost mainly from the full model, not from the reference models.

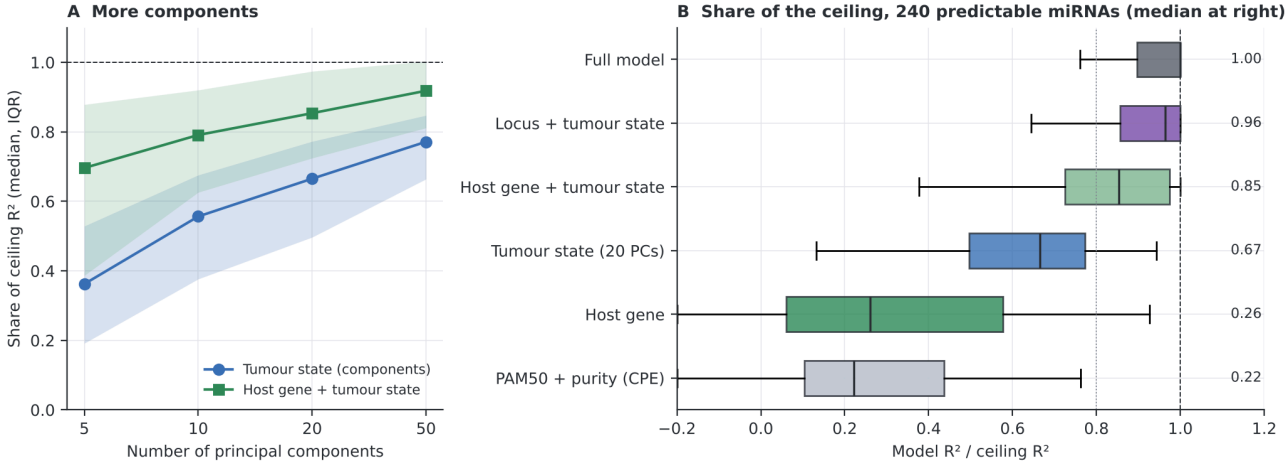


Figure 4. Share of the ceiling R^2 reproduced by each model, for the 240 miRNAs with ceiling $R^2 \geq 0.3$ (plate-grouped folds). (A) Tumour state (ridge on the leading principal components) with and without the host gene, as the number of components increases; line, median; band, interquartile range. (B) Ratio of each model's R^2 to the ceiling; boxes, interquartile range; whiskers, 1.5 × IQR; dotted line, 0.8, the threshold used for attribution. The host-gene model is shown for the 167 predictable miRNAs with a measurable host gene.

The leading components reflect tumour biology, with one batch-related axis

The leading components correspond to familiar axes of breast-tumour biology (Table 3). The first two separate ER/luminal from other tumours; PC2 is the component most associated with PAM50 subtype ($\eta^2 = 0.44$) and also correlates with purity ($\rho = 0.45$). PC4 contrasts proliferation with endothelial and fibroblast content and is the component most associated with purity ($\rho = 0.54$), and PC5 and PC6 capture fibroblast and lymphocyte content. PC3 is different. It correlates with basal and T-cell programmes, but sequencing plate explains 0.47 of its variance (0.45 adjusted), compared with at most 0.21 for the other nine components; it is also associated with tissue source site ($\eta^2 = 0.20$), RNA-seq library size ($\rho = 0.27$) and the number of miRNA-mapped reads ($\rho = -0.33$). One of the leading components is therefore mainly a batch axis. These interpretations come from correlations with marker-gene programme scores (Supplementary Table S3) and with technical variables, and are descriptive.

Two analyses tested whether the component model depends on batch or on host and local genes. First, plate effects were removed from both the mRNA and the miRNA data within each training fold, and all models were refitted. This used patient-grouped folds, because a plate effect can only be estimated for plates seen in training. The 20 components then reproduced a median of 66% of the ceiling (unadjusted 64%), the components with the host gene(s) 85% (79%) and with the locus genes 98% (89%). The predictive value of the components is therefore not mainly explained by plate. Second, rebuilding the components without the 4,080 host genes and genes within ±500 kb of any target changed only 4 labels (Table 5). "Tumour state" is used below as shorthand for the component-based reference model, with the caveat that one of its components is batch-related.

Table 3. The ten leading principal components of tumour gene expression (1,053 tumours, all data) with the two marker-gene programmes most correlated with each (Spearman ρ; the sign of a component is arbitrary) and their association with subtype, purity and technical variables. η^2, share of the component's variance explained by PAM50 subtype, mRNA sequencing plate (40 plates) or tissue source site (40 sites), with the value adjusted for the number of groups in brackets; ρ, Spearman correlation with CPE purity, RNA-seq library size and the number of miRNA-mapped reads.

PC	Var.	Leading programmes (ρ)	PAM50 η^2	CPE ρ	Plate η^2 (adj.)	Site η^2 (adj.)	Library ρ	miRNA reads ρ
PC1	12.9%	ER/luminal (0.60); ciliary (0.50)	0.17	0.16	0.17 (0.14)	0.12 (0.09)	0.20	0.10
PC2	6.4%	ER/luminal (0.57); ciliary (0.42)	0.44	0.45	0.21 (0.18)	0.12 (0.08)	−0.21	0.24
PC3	5.2%	basal (−0.50); T cell (−0.42)	0.19	0.33	0.47 (0.45)	0.20 (0.17)	0.27	−0.33
PC4	4.7%	proliferation (0.67); endothelial (−0.62)	0.29	0.54	0.08 (0.04)	0.07 (0.03)	0.06	−0.23
PC5	3.3%	fibroblast (0.48); myeloid (0.28)	0.07	−0.27	0.13 (0.09)	0.08 (0.04)	−0.02	−0.01
PC6	2.2%	T cell (0.47); B/plasma (0.41)	0.17	−0.26	0.05 (0.01)	0.09 (0.06)	0.06	−0.04
PC7	1.6%	fibroblast (−0.27); T cell (0.26)	0.03	−0.04	0.17 (0.14)	0.09 (0.05)	0.09	−0.04
PC8	1.5%	fibroblast (−0.24); endothelial (0.18)	0.09	0.03	0.05 (0.01)	0.04 (0.01)	−0.03	0.01
PC9	1.3%	fibroblast (0.28); epithelial (0.12)	0.07	−0.06	0.08 (0.04)	0.05 (0.01)	0.06	0.05
PC10	1.1%	myeloid (0.33); B/plasma (−0.26)	0.12	−0.05	0.08 (0.04)	0.08 (0.05)	0.09	0.00

Attribution of each miRNA

Each predictable miRNA was assigned to the first reference model, in a fixed order, that reproduced at least 80% of its ceiling R^2 (Table 4): the host gene(s) alone, tumour state alone, the host gene(s) with tumour state, and the genomic locus with tumour state, sub-labelled own or paralog locus. miRNAs whose full-model R^2 exceeded the best of all reference models by at least 0.10 were labelled residual signal, and the remainder partly explained. Of the 240 predictable miRNAs, 19 were assigned to the host gene, 39 to tumour state, 85 to the host gene with tumour state, 60 to the own locus and 5 to the paralog locus, while 25 retained residual signal and 7 were partly explained (Figure 5). For 208

(87%) of the predictable miRNAs, a reference model built from host, local or component information was sufficient. Compared with patient-grouped folds, 79 labels changed: residual signal fell from 47 to 25, partly explained from 24 to 7, and tumour state rose from 29 to 39. Results for every miRNA are listed in Supplementary Table S1.

Table 4. Attribution categories, applied in the order listed, with the number of miRNAs under plate-grouped (main) and patient-grouped folds. R^2 values are cross-validated means; "ceiling" is the highest R^2 of the full model or any reference model; "full" is the ridge model on all genes; "best reference" is the highest R^2 among the reference models. Examples are the miRNAs with the highest ceiling R^2 in each category.

Category	Rule (applied in this order)	Plate	Patient	Highest ceiling R^2
Poorly predicted	ceiling $R^2 < 0.3$	97	54	–
Host gene	host gene(s) alone ≥ 0.8 × ceiling	19	18	mir-30a, mir-6510, mir-224, mir-452
Tumour state	20 principal components ≥ 0.8 × ceiling	39	29	mir-17, mir-150, mir-190b, mir-18a
Host gene + tumour state	host gene(s) + 20 components ≥ 0.8 × ceiling	85	90	mir-10a, mir-4728, mir-210, mir-140
Own locus	genes within ±500 kb + 20 components ≥ 0.8 × ceiling; own-locus genes predict better than paralog-locus genes	60	68	mir-196a-2, mir-149, mir-3614, mir-375
Paralog locus	as above, but genes at the paralog's locus predict better	5	7	mir-196a-1, mir-9-2, mir-199a-1, mir-218-2
Residual signal	full − best reference ≥ 0.10	25	47	mir-3200, mir-203a, let-7b, mir-203b
Partly explained	none of the above	7	24	mir-23a, mir-199b, mir-1468, mir-940

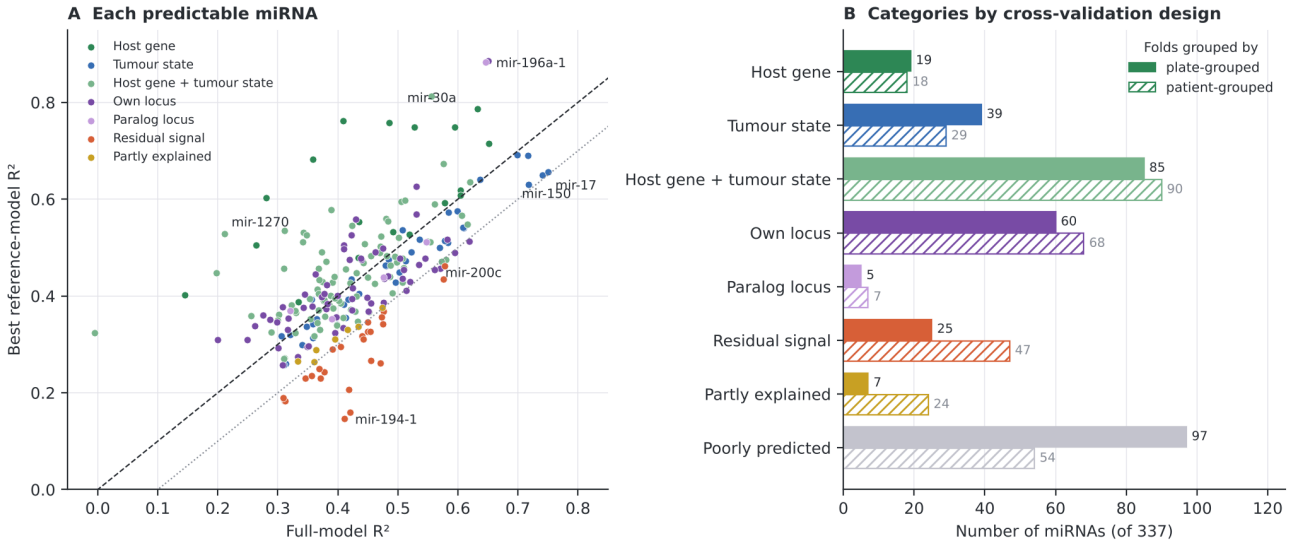


Figure 5. Attribution of the predictable miRNAs. (A) Full-model R^2 against the best reference-model R^2 (plate-grouped folds); dashed line, equal R^2; dotted line, a gap of 0.10, below which miRNAs not matched by a reference model are labelled residual signal. Points above the dashed line are predicted better by a reference model than by all genes. (B) Number of miRNAs in each category with plate-grouped (filled) and patient-grouped (hatched) folds.

Tumour-state miRNAs followed a few programmes. Proliferation was the most common (14 miRNAs; for example mir-92a-1, mir-92a-2, mir-130b and mir-106b, the last encoded in an intron of the DNA replication gene MCM7), followed by the ER/luminal axis (12: mir-17 and mir-18a of the miR-17~92 cluster and mir-9-3 were higher in ER-low tumours, and mir-190b was higher in ER/luminal tumours), fibroblast content (7; including mir-379, mir-337 and mir-299 of the

DLK1-DIO3 cluster described below) and T-cell content (4; mir-150, mir-155, mir-146a and mir-142); 2 followed other programmes. For mir-150, which has no host gene, 20 components reproduced 85% of its ceiling of 0.74, and its level correlated with the T-cell programme at ρ = 0.80. This is consistent with miR-150 marking lymphocyte content rather than a tumour-cell property (Xiao et al., 2007).

Robustness and uncertainty of the labels

Three analyses tested how much the attribution depends on analysis choices and on fold-to-fold variation. First, the number of components, the locus window and the sufficiency threshold were varied (Table 5). The number of components had a large effect: with 10 components, 62 labels changed, residual signal grew from 25 to 48 and tumour state fell from 39 to 15; with 50 components, 78 changed, tumour state grew to 84 and residual signal fell to 5. The threshold had a similar effect: at 0.7, 86 labels changed and tumour state grew to 94; at 0.9, 100 changed, host-gene and tumour-state labels almost disappeared (2 and 7) and 55 miRNAs became partly explained. Halving or doubling the locus window changed 6 and 11 labels, and rebuilding the components without host and local genes changed 4. Second, the uncertainty of the paired difference that set each label was estimated across the 15 test folds with the corrected resampled t interval of Nadeau and Bengio (2003), which allows for the overlap between training sets. For 131 of the 240 predictable miRNAs the 95% interval crossed the threshold (Table 6), as did 60 of the 97 poorly predicted miRNAs for the 0.3 cut-off. The interval is conservative, so these numbers are an upper bound on how many labels are uncertain.

The aggregate results are robust: in every alternative setting of Table 5 a simpler model was sufficient for most predictable miRNAs (from 64% to 95%), and the median shares of Figure 4 changed gradually with the number of components rather than abruptly. The label of an individual miRNA is less certain. 104 of 240 labels were identical across locus windows and gene exclusion and clear of the threshold, and 84 were also identical with 10, 20 and 50 components (with patient-grouped folds, 120 of 283 and 96). These form the most reliable part of Supplementary Table S1, where each miRNA is marked accordingly. The robust core includes host-gene miRNAs (mir-6510, mir-224, mir-205, mir-135b), tumour-state miRNAs (mir-18a, mir-155, mir-106b, mir-590), own-locus miRNAs (mir-196a-2, mir-148a, mir-542, mir-200a, mir-200b, mir-758) and host-gene + tumour-state miRNAs (mir-4728, mir-615, mir-342, mir-532). No partly explained label was robust.

Table 5. Number of miRNAs in each category under alternative settings (plate-grouped folds; residual gap 0.10 in all settings). Changed, number of labels, among all 337, that differ from the main analysis.

Setting	Host	State	Host + state	Own locus	Paralog locus	Residual	Partly	Changed
Main analysis (20 components, ±500 kb, 0.8)	19	39	85	60	5	25	7	–
10 components	21	15	83	54	7	48	11	62
50 components	15	84	79	52	3	5	6	78
Locus window ±250 kb	19	38	85	59	5	27	7	6
Locus window ±1 Mb	20	41	84	55	5	26	8	11
Components without host and local genes	19	38	86	59	5	25	8	4
Sufficiency threshold 0.7	29	94	61	40	3	13	0	86
Sufficiency threshold 0.9	2	7	84	51	9	32	55	100

Table 6. Uncertainty and robustness of the labels by category. Borderline, the 95% corrected resampled t interval of the deciding difference crosses the threshold. All folds, the deciding condition holds in all 15 test folds. Robust, the same label with locus windows of ±250 kb, ±500 kb and ±1 Mb and with components rebuilt without host and local genes, and not borderline; incl. components, also the same with 10, 20 and 50 components.

Category	miRNAs	Borderline	All folds	Robust	Robust incl. components
Host gene	19	15	2	4	4
Tumour state	39	30	4	9	5
Host gene + tumour state	85	32	49	52	43
Own locus	60	25	31	31	25
Paralog locus	5	2	3	3	2
Residual signal	25	20	4	5	5
Partly explained	7	7	0	0	0
Poorly predicted	97	60	31	37	37
All	337	191	124	141	121

Host genes and genomic loci

Of the 337 targets, 269 have a host gene (137 in a protein-coding intron, 20 in a protein-coding exon, 108 in a non-coding host transcript and 4 in a transcribed pseudogene); 66 are intergenic and two could not be mapped. For 45 miRNAs the host gene was too weakly expressed in the poly-A RNA-seq data to be used (Methods). For the 19 host-gene miRNAs, a linear model on the host gene(s) reproduced at least 80% of the ceiling and often matched or exceeded the full model on all 27,671 genes: for example mir-135b (host genes LEMD1 and BLACAT1; 0.66 vs 0.65), mir-452 and mir-224 (both in GABRE; 0.65 vs 0.59 and 0.64 vs 0.49), mir-30a (LINC00472 and AL136164.2; 0.66 vs 0.63), mir-205 (MIR205HG; 0.63 vs 0.53) and mir-6510 (KRT15; 0.71 vs 0.41; host R^2 versus full R^2). Combining the host gene with the components was sufficient for a further 85 miRNAs, including several that the full model predicted poorly: mir-1270 (full R^2 0.21, host gene + components 0.53) and mir-675, which lies in H19 (0.20 versus 0.45). The same applied to four miRNAs that fell below the expression threshold because they are absent from many tumours (Supplementary Table S2). These include mir-934 (VGLL1; host R^2 0.89 versus 0.79) and mir-449a (CDC20B; 0.85 versus 0.69). VGLL1 and miR-934 are expressed mainly in basal-like tumours (Castilla et al., 2014), and the miR-449 cluster lies in an intron of CDC20B, which is co-expressed with the multiciliogenesis regulators MCIDAS and CCNO (Revinski et al., 2018).

For the 60 own-locus miRNAs, the annotated host gene was weak or absent, but genes within ±500 kb together with tumour state reproduced at least 80% of the ceiling. The gain from the local genes over tumour state alone was a median of 0.11 R^2 units across the 65 locus miRNAs. Several are known lineage miRNAs whose primary transcripts are poorly annotated: the epithelial miR-200 family (Gregory et al., 2008), with mir-200c and mir-141 predicted by own-locus genes at R^2 0.40 and 0.43 although the annotated host MIR200CHG reached only 0.10, and mir-200a, mir-200b and mir-429 at 0.36–0.47; mir-375 (0.43); and mir-3065 (0.36). Another large group lies in the imprinted DLK1-DIO3 region on chromosome 14, which encodes one of the largest miRNA clusters in the genome within the MEG3–MEG8 non-coding transcript (Benetatos et al., 2013). Of the 25 DLK1-DIO3 targets, 19 were attributed to the own locus, 3 to tumour state (all following the fibroblast programme) and 3 were poorly predicted, which is consistent with, but does not show, expression of this cluster mainly in stromal cells. At HOX clusters the neighbourhood mattered most: mir-10a was predicted from its host HOXB3 at R^2 0.62 and from its locus at 0.79, compared with 0.56 for the full model. Figure 6 compares the reference models for representative miRNAs of each category.

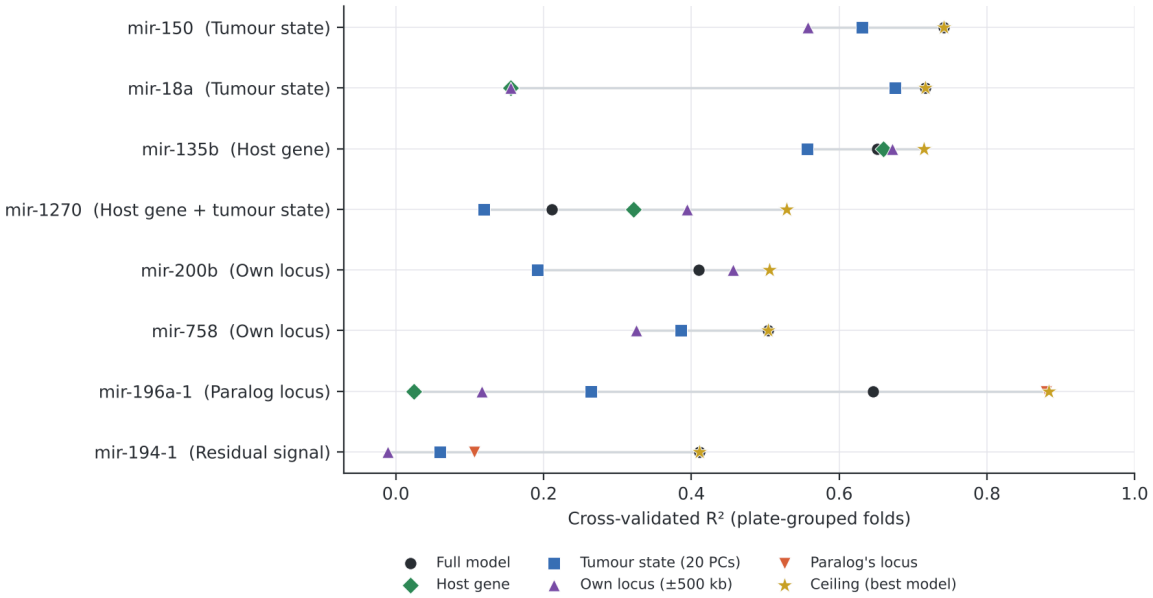


Figure 6. Cross-validated R^2 of the reference models for representative miRNAs (plate-grouped folds), with each miRNA's attribution category in brackets. Own and paralog locus: ridge on genes within ±500 kb of the miRNA or of its paralogs, without tumour state. Ceiling: the highest R^2 of the full model or any reference model. mir-150, mir-200b, mir-758 and mir-194-1 have no measurable host gene.

Paralogous precursors are often better predicted by each other's locus

Paralogous precursors, such as mir-196a-1 (in the HOXB cluster) and mir-196a-2 (in the HOXC cluster), produce identical or near-identical mature miRNAs from different genomic loci. For 17 of the 32 paralogous precursors with full-model $R^2 \geq 0.3$, genes at the paralog's locus predicted the precursor better than genes at its own locus (Figure 7). For the 44 paralogous precursors with $R^2 \geq 0.3$ under patient-grouped folds, the result was 25 of 44 under both designs, and no precursor changed direction. The clearest case was mir-196a-1: genes at its own HOXB locus reached R^2 0.12, whereas genes at the HOXC locus of mir-196a-2 reached 0.88, practically the same value as for mir-196a-2 itself (0.88). Similar patterns were seen for mir-9-2 (0.02 versus 0.47) and mir-550a-1 (0.15 versus 0.29). These patterns are consistent with ambiguity in precursor-level quantification and caution against interpreting the reported values as locus-specific expression. One possible explanation is that reads from the shared mature sequence contribute to more than one precursor; this was not tested against the quantification workflow or read-level data and remains a hypothesis.

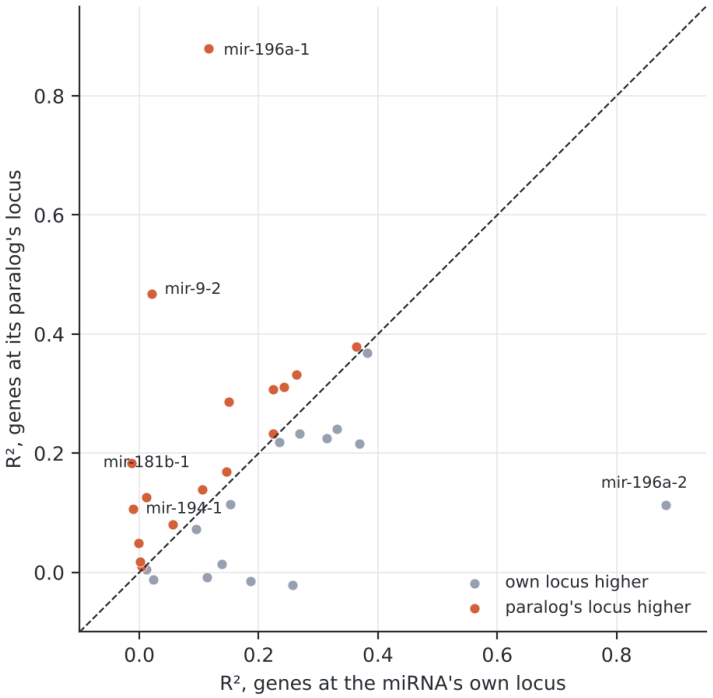


Figure 7. Prediction of 32 paralogous miRNA precursors (full-model $R^2 \geq 0.3$, plate-grouped folds) from genes within ±500 kb of their own locus versus genes at the loci of their paralogs. Orange, the paralog's locus predicts better; dashed line, equal R^2.

Residual predictive signal beyond the reference models

25 miRNAs were predicted at least 0.10 R^2 units better by all genes than by any of the reference models (median gap 0.13; Table 7). All of them also had residual signal under patient-grouped folds, but 22 of the 47 residual-signal miRNAs under patient-grouped folds lost it when plates were held out: 9 became poorly predicted (including mir-143, mir-326, mir-330 and mir-15a), 5 were matched by the host gene with tumour state, 4 by the locus and 4 became partly explained. About half of the apparent miRNA-specific signal therefore did not transfer to unseen batches. The remaining group should also be interpreted with care. 11 of the 25 carry at least one quantification or annotation flag: 9 are paralogs whose two loci each explain part of the signal (mir-194-1 and mir-194-2, mir-181a-1, mir-181b-1 and mir-181b-2, mir-16-1 and mir-16-2, mir-103a-1 and mir-103a-2), and 4 have reads flagged as cross-mapped to other miRNAs. Only 5 residual-signal labels were robust, and all of them belong to the mir-181 and mir-194 paralog groups. The other residual-signal miRNAs, including mir-203a, mir-484, mir-146b and mir-184, carry predictive signal that is not captured by the host gene(s), the ±500 kb neighbourhood or 20 linear components, but their labels are borderline. This does not mean they are unexplained by all known factors, and the analysis does not establish what the additional signal represents.

Table 7. The 25 residual-signal miRNAs, sorted by the gap between the full model and the best reference model (plate-grouped folds). Full R^2, patient: the full model with patient-grouped folds. Robust: ✓, robust including the number of components; (✓), robust except for the number of components; b, borderline. Flags: X, reads cross-mapped in more than half of the samples; P, member of a paralog group; H, host gene not measurable.

miRNA	Full R^2	Full R^2, patient	Best reference R^2	Gap	Robust	Host gene(s)	Flags
mir-194-1	0.41	0.44	0.15	0.26	✓	none	P
mir-194-2	0.42	0.46	0.16	0.26	✓	MIR194-2HG	P H
mir-181a-1	0.42	0.45	0.21	0.21	✓	MIR181A1HG	P H
mir-181b-1	0.47	0.49	0.26	0.21	✓	MIR181A1HG	P H
mir-181b-2	0.46	0.48	0.27	0.19	✓	MIR181A2HG	P

miRNA	Full R^2	Full R^2, patient	Best reference R^2	Gap	Robust	Host gene(s)	Flags
mir-107	0.37	0.42	0.23	0.14	b	PANK1	X
mir-203a	0.58	0.60	0.43	0.14	b	none	
mir-192	0.38	0.46	0.24	0.13	b	MIR194-2HG	X H
mir-203b	0.48	0.51	0.34	0.13	b	none	
mir-484	0.44	0.51	0.31	0.13	b	NDE1	
mir-103a-2	0.31	0.47	0.18	0.13	b	PANK2	X P
mir-103a-1	0.31	0.47	0.18	0.13	b	PANK3	X P
mir-16-2	0.45	0.54	0.33	0.13	b	SMC4	P
mir-16-1	0.45	0.53	0.33	0.12	b	DLEU2	P
mir-184	0.44	0.46	0.32	0.12	b	none	
mir-760	0.36	0.44	0.23	0.12	b	none	
mir-425	0.37	0.43	0.25	0.12	b	DALRD3	
mir-374a	0.31	0.38	0.19	0.12	b	FTX	
mir-3200	0.58	0.60	0.46	0.12	b	OSBP2	
mir-132	0.35	0.40	0.23	0.12	b	none	
mir-146b	0.47	0.57	0.36	0.12	b	none	
mir-769	0.40	0.47	0.30	0.11	b	CCDC61	
let-7b	0.48	0.54	0.37	0.11	b	MIRLET7BHG	
mir-1266	0.45	0.50	0.35	0.10	b	MYO5C	
mir-98	0.39	0.40	0.29	0.10	b	HUWE1	

Discussion

This analysis asked what mRNA-based predictions of miRNA expression capture in breast tumours. Most well-expressed miRNA precursors could be predicted from mRNA even when whole sequencing batches were held out (240 of 337 with $R^2 \geq 0.3$), but part of the accuracy measured with the more common patient-grouped design came from batch structure shared by the two assays, and for most miRNAs a much simpler reference model reproduced most of what remained. The findings are discussed in terms of batch effects, which proxies are sufficient, what this means for the use of inferred miRNA levels, and what the analysis cannot resolve.

Shared batches. In TCGA-BRCA the mRNA and miRNA libraries of a tumour were sequenced on corresponding plates, so a model trained on mRNA can learn a plate's signature and transfer it to the miRNA of a test tumour from the same plate. Holding out plates lowered the full model's R^2 by about 11%, three times as much as holding out tissue source sites, and removed about half of the miRNAs that appeared to carry signal beyond all simpler models. This is an instance of the bias described by Soneson et al. (2014): when batches are shared between features and outcome and are not held out, cross-validation overestimates performance. The full model was affected more than the component model, which suggests that a flexible model with many genes can pick up subtle batch signatures that the leading components summarise only partly. Models that are trained on TCGA and evaluated on held-out TCGA samples without grouping by batch may therefore report optimistic accuracy, and the miRNAs they appear to predict best beyond simple baselines are the ones most likely to be affected.

Which proxies are sufficient. Three kinds of simpler model were sufficient for most predictable miRNAs. These comparisons show predictive sufficiency; they do not partition expression into causes, because the reference models overlap in the information they use. The first kind is the host gene or the surrounding locus. For intronic and lncRNA-hosted miRNAs the host gene(s) often predicted the miRNA as well as the whole transcriptome, consistent with co-transcription (Rodriguez et al., 2004; Baskerville & Bartel, 2005). Where the annotated host was a poor proxy, as for the miR-200 family, mir-375 and the DLK1-DIO3 cluster, nearby genes were sufficient instead, possibly because they track the same poorly annotated primary transcripts or the chromatin state of the locus. The second kind is the low-dimensional structure of the transcriptome. Twenty principal components, which correlate with ER/luminal, basal, proliferation, fibroblast, endothelial and lymphocyte programmes, reproduced two-thirds of the transferable predictive performance for a typical miRNA, whereas explicit covariates reproduced about a fifth. Subtype labels and purity estimates are therefore not adequate substitutes for this structure. One component mainly reflected sequencing plate, but removing plate effects from both data types did not reduce the components' share. Lymphocyte miRNAs such as miR-150, miR-155 and miR-142 behaved as markers of immune content, mir-17 and mir-18a as markers of the ER-low state, and several miR-17~92 and MCM7-hosted miRNAs as markers of proliferation. The third observation concerns quantification: paralogous precursors were often better predicted by each other's loci, which suggests that their precursor-level values are not locus-specific, and the most robust residual-signal miRNAs were all paralogs.

Implications for inferred miRNA levels. Tools that infer miRNA expression from RNA-seq are attractive because miRNA sequencing is missing from most transcriptomic datasets (Olgun et al., 2022; Cai et al., 2025). The results here suggest that, in bulk breast tumours, an inferred miRNA level may often act as a stand-in for its host transcript, for the cell composition of the sample or for the tumour's subtype and proliferation state, and that part of a model's apparent accuracy can come from batch. Such a value can still be useful, but it should not be read as evidence of miRNA-specific regulation unless it adds information beyond these baselines. Two practical checks follow from this study: evaluate inference with folds that hold out whole processing batches, and compare it with host-gene, locus and tumour-state reference models fitted on the same folds. The per-miRNA results of this study (Supplementary Table S1) can serve as such a reference for breast tumours, preferably restricted to the labels marked robust, and within the limits described

below. In an exploratory comparison, a regularised linear model also outperformed a default random forest for every miRNA tested, and SHAP rankings from the random forest were too unstable to support gene-level interpretation (Supplementary Figure S1), in agreement with the pan-cancer comparison of Cihan et al. (2025).

Limitations. First, the plate-grouped design was adopted during the analysis, after the batch structure was found, and several other choices (the ceiling, the host-measurability filter and the robustness criteria) were also refined along the way; results under patient-grouped folds are reported alongside so that the effect of these choices is visible. Second, plate and tissue source site overlap, and plates may also differ in subtype mix or collection period, so the loss under plate-grouped folds may include some biological differences between plates as well as technical effects; the plate adjustment likewise removes some biology that varies between plates. Third, the attribution categories depend on choices: the 0.8 and 0.10 thresholds, the ±500 kb window and above all the number of principal components (Table 5), and with more components the component model reproduces more of the prediction (Figure 4A), so the boundary between categories is a matter of degree. The categories are descriptive summaries of predictive sufficiency, not mechanisms. Fourth, the ceiling is the maximum of seven models evaluated on the same folds and is therefore slightly optimistic, and the paired differences that set the labels are uncertain: a corrected resampled t interval placed 131 of 240 labels within reach of the threshold. Fifth, "tumour state" is an interpretation of the component model rather than a measured quantity, and one of its components is batch-related. Sixth, the data are bulk, poly-A-selected RNA-seq, in which primary miRNA transcripts and some non-coding hosts are poorly captured, so host-gene R^2 is a lower bound; miRNA values are precursor-level RPM, not mature or isoform-level counts. Seventh, LUMP purity was missing for a third of tumours. Eighth, all results come from one cohort and one sequencing platform, and transfer to independent cohorts was not tested. Finally, the analysis is correlational. It identifies what a miRNA's level co-varies with but does not test whether the miRNA represses its targets.

Future work. The most direct extension is to apply the same batch-aware evaluation to other TCGA cancer types and to retrain a published miRNA-inference model with random and with plate-grouped splits, to measure how much its reported accuracy depends on shared batches. The miRNAs with residual signal are a starting point for testing function, for example by asking whether their predicted targets (from TargetScan or validated interaction databases) are expressed lower than expected once host, locus and component-based structure are accounted for. Isoform-level reads, which carry genomic coordinates, could test whether paralogous precursors are quantified locus-specifically, and an external cohort with paired data would test how general these results are.

Conclusion

In TCGA breast tumours, mRNA expression predicted most well-expressed miRNA precursors even when whole sequencing batches were held out, but part of the accuracy measured with patient-grouped cross-validation came from batches shared by the two assays, and simpler models using host genes, genomic neighbourhoods and low-dimensional transcriptomic structure reproduced most of the rest. These comparisons identify sufficient predictive proxies rather than causal mechanisms, and the label of any single miRNA is less certain than the aggregate pattern. Paralog-associated patterns also caution against locus-specific interpretation of precursor-level measurements. Inferred miRNA expression should therefore be evaluated with batch-aware cross-validation and against these baselines before being interpreted as evidence of miRNA-specific biology.

Materials and methods

The workflow is summarised in Figure 1.

Data

Open-access TCGA-BRCA expression files were downloaded from the Genomic Data Commons (Grossman et al., 2016) with gdc-client, using a manifest and sample sheet generated on 17 August 2026. Gene expression was taken from the STAR "augmented gene counts" files (column tpm_unstranded; 60,660 genes) and miRNA expression from the miRBase 21 miRNA quantification files (column reads_per_million_miRNA_mapped, 1,881 precursors; Kozomara et al., 2019), together with the per-file cross-mapped flag. Primary tumours (tissue type "Tumor", descriptor "Primary") with both data types were matched by sample ID (1,087 tumours). Five sample IDs had two files of each data type, all from tissue source site A7, where the same patients also had an FFPE sample. The first file listed in the sample sheet was used; for mRNA this was in every case the file less similar to the patient's FFPE profile and more similar to other tumours, and the file IDs used are recorded in the repository. The aliquot barcodes of the files used were retrieved from the GDC API (one metadata query).

Cohort

Samples recorded as formalin-fixed paraffin-embedded in the sample sheet (13) were removed, which left one tumour per patient (1,074). PAM50 subtype calls (TCGAquery_subtype) and tumour purity estimates (Tumor.purity: CPE and LUMP; Aran et al., 2015) were obtained from the TCGAbiolinks package (Colaprico et al., 2016), matched by patient and sample barcode respectively. Tumours with a PAM50 call and a CPE value formed the analysis set (1,053; Table 1).

Targets and features

Expression values were transformed as $\log2(x + 1)$. Targets were miRNA precursors with RPM > 1 in at least 50% of the analysis set (337); features were genes with TPM > 0.5 in at least 10% of the analysis set (27,671). Neither selection used information about the association between genes and miRNAs.

Batches, sites and cross-validation designs

The sequencing plate of each file was taken from the fifth field of its aliquot barcode, and the tissue source site from characters 6–7 of the sample barcode. The association between mRNA plate, miRNA plate and site was summarised with Cramér's V (with and without bias correction). In the main design, folds were formed by assigning whole mRNA plates to five folds balanced by tumour count (211, 211, 211, 210 and 210 tumours), repeated with seeds 42, 43 and 44 to give 15 test folds; because mRNA and miRNA plates were nearly interchangeable, this also held out the miRNA plates. For comparison, the same models were fitted with 5-fold cross-validation grouped by patient (GroupKFold with shuffling, seeds 42–44), and the full and 20-component models with folds of whole tissue source sites balanced by tumour count (3 seeds). R^2 was computed per test fold and summarised as the mean and SD over the 15 folds; for the comparison of designs, the pooled out-of-fold R^2 of each repeat (one prediction per tumour) was also computed and averaged over repeats.

Models

Ridge regression (Hoerl & Kennard, 1970) was fitted with RidgeCV in scikit-learn (Pedregosa et al., 2011), with features standardised on each outer training fold and the penalty chosen per miRNA from 41 log-spaced values between 10^{-2}

and 10^8 by efficient leave-one-out cross-validation within that training fold. The standardisation was fitted once per outer training fold and held fixed during the internal penalty selection, so it can influence the choice of penalty but not the scoring of the outer test fold. The following models were fitted for each miRNA:

- **Full model:** ridge regression on all 27,671 genes.
- **Host gene:** linear regression on the log2 TPM of the host gene(s), fitted only if at least one host gene passed the feature filter (host gene measurable); otherwise host-gene models were not fitted and host + component models reduce to the component models.
- **Tumour state:** principal component analysis (Pearson, 1901) of the standardised gene matrix within each training fold, with test tumours projected onto the components; ridge regression on the leading k components (k = 5, 10, 20 and 50; k = 20 for attribution), with and without the host gene(s).
- **Genomic locus:** ridge regression on the filtered genes within ±500 kb of the miRNA precursor (own locus), within ±500 kb of the other members of its paralog group (paralog locus), on their union, and on the union with 20 components, with and without the host gene(s).
- **Clinical covariates:** linear regression on PAM50 subtype (one-hot, Basal as reference) and CPE purity, and on the same plus LUMP purity with missing values (348 tumours) set to the training-fold mean and a missingness indicator.

Host genes, loci and paralogs

Precursors were mapped to their MIR genes in GENCODE v36 (Frankish et al., 2021), the annotation used by the GDC, by name (for example hsa-mir-449a to MIR449A, hsa-let-7a-1 to MIRLET7A1); two precursors could not be mapped. Host genes were non-miRNA genes overlapping the precursor on the same strand and were classified as protein-coding intron, protein-coding exon, non-coding host (including MIR…HG transcripts) or transcribed pseudogene. The measurability filter was introduced because a host gene that is essentially absent from the data gives an unstable one-gene regression. For example, MEG8, the annotated host of eight DLK1-DIO3 miRNAs, exceeded 0.5 TPM in only one tumour (9.1 TPM; the next highest was 0.27). That tumour had a leverage of 0.92–0.95 in the training folds that contained it, and in the folds where it was a test tumour it lay more than 100 SD from the training mean, which gave host-gene R^2 values of about −1 for these miRNAs. Paralog groups were precursors whose names differ only in a numeric suffix (for example mir-196a-1 and mir-196a-2).

Attribution

For each miRNA, the ceiling was the model with the highest mean R^2 among the full model and six reference models (host gene(s); 20 components with and without host gene(s); locus genes; locus genes with 20 components with and without host gene(s)), and its mean R^2 was the ceiling R^2. miRNAs with ceiling $R^2 < 0.3$ were labelled poorly predicted. The others were assigned to the first category, in this order, whose condition was met: host gene (host-gene $R^2 \geq 0.8 \times$ ceiling); tumour state (20-component $R^2 \geq 0.8 \times$ ceiling); host gene + tumour state (host gene(s) + 20 components $\geq 0.8 \times$ ceiling); locus (locus genes + 20 components $\geq 0.8 \times$ ceiling; own locus or paralog locus by whichever of the two locus-only models had the higher R^2); residual signal (full R^2 minus the highest R^2 among the six reference models $\geq$ 0.10); and partly explained otherwise. For tumour-state miRNAs, the programme with the largest absolute Spearman correlation with the miRNA was reported.

Robustness and uncertainty

The attribution was repeated with 10 and 50 components (including the locus models with that number of components), with locus windows of ±250 kb and ±1 Mb, with sufficiency thresholds of 0.7 and 0.9, and with components

rebuilt after removing all host genes of the 337 targets and all genes within ±500 kb of any target (4,080 genes); the ceiling was recomputed in each setting. For uncertainty, the per-fold difference that decided each label (the reference model's R^2 minus 0.8 × the ceiling model's R^2 in the same fold; for residual signal, the gap minus 0.10; for poorly predicted, 0.3 minus the ceiling model's R^2) was summarised across the 15 test folds with the corrected resampled t interval of Nadeau and Bengio (2003; test-to-training ratio 0.25, 14 degrees of freedom). A label was borderline if this 95% interval included zero; partly explained labels were borderline if the interval of either the closest sufficiency condition or the residual condition reached zero. A label was robust if it was identical with locus windows of ±250 kb, ±500 kb and ±1 Mb and with the rebuilt components and was not borderline, and robust including components if it was also identical with 10, 20 and 50 components.

Plate adjustment

To test whether the component model depends on plate, the mean of each gene and each miRNA within each plate (minus the training-fold grand mean) was subtracted within each training fold of the patient-grouped design, the same training-fold shifts were applied to the test tumours, and all models, including the locus models, were refitted on the adjusted data. R^2 values from this analysis refer to the adjusted miRNA values.

Programme scores and technical variables

Twelve marker-gene programmes (Supplementary Table S3) were scored as the mean z-score of log2(TPM + 1) of their genes across the analysis set. They were used only to interpret miRNAs and principal components, not in any model. For the ten leading principal components of the full analysis set, the variance explained by PAM50 subtype, mRNA plate, miRNA plate and tissue source site was summarised as η^2 (with the adjustment for the number of groups), and Spearman correlations were computed with CPE purity, RNA-seq library size (the sum of unstranded counts in the STAR file) and the number of miRNA-mapped reads (the sum of read counts in the miRNA file).

Software

Analyses used Python 3.12 with NumPy 2.4.6, pandas 3.0.5, SciPy 1.18.0, scikit-learn 1.9.0, shap 0.52.0 and matplotlib 3.11.1, and R 4.5.3 with TCGAbiolinks 2.38.0. Random seeds were fixed throughout.

Supplementary section

Supplementary Table S1. Attribution of all 337 miRNA precursors, sorted by ceiling R^2 (plate-grouped folds unless stated). Full, mean (SD) over 15 test folds; Patient, full model with patient-grouped folds; Host, host-gene model; State, 20 principal components; Ceiling, the highest R^2 of the full model or any reference model. Robust: ✓, robust including the number of components; (✓), robust except for the number of components; b, borderline (95% interval crosses the threshold); s, label changes with the locus window or with the rebuilt components. Flags: X, cross-mapped reads; P, paralog group; H, host gene not measurable. A machine-readable version with all model results is in the repository.

miRNA	Full (SD)	Patient	Host	State	Ceiling	Category	Robust	Host gene(s)	Flags
mir-196a-2	0.65 (0.04)	0.66	0.14	0.26	0.89	Own locus	✓	AC012531.3;HOXC6	P
mir-196a-1	0.65 (0.04)	0.65	0.02	0.26	0.88	Paralog locus	✓	HOXB7	P
mir-10a	0.56 (0.04)	0.57	0.62	0.19	0.81	Host gene + tumour state	b	HOXB3;AC103702.1	
mir-30a	0.63 (0.05)	0.65	0.66	0.35	0.79	Host gene	b	AL136164.2;LINC00472	X
mir-6510	0.41 (0.07)	0.42	0.71	0.24	0.76	Host gene	✓	KRT15	
mir-224	0.49 (0.07)	0.52	0.64	0.38	0.76	Host gene	✓	GABRE	
mir-17	0.75 (0.06)	0.80	0.23	0.63	0.75	Tumour state: ER/luminal (−)	b	MIR17HG	X
mir-205	0.53 (0.05)	0.54	0.63	0.30	0.75	Host gene	✓	MIR205HG	
mir-452	0.59 (0.04)	0.62	0.65	0.46	0.75	Host gene	b	GABRE	
mir-150	0.74 (0.06)	0.77	–	0.63	0.74	Tumour state: T cell (+)	(✓)		
mir-190b	0.72 (0.04)	0.75	−0.02	0.58	0.72	Tumour state: ER/luminal (+)	b	TPM3	
mir-18a	0.72 (0.05)	0.74	0.16	0.68	0.72	Tumour state: ER/luminal (−)	✓	MIR17HG	
mir-135b	0.65 (0.06)	0.67	0.66	0.56	0.71	Host gene	✓	LEMD1;BLACAT1	
mir-155	0.70 (0.05)	0.72	0.50	0.66	0.70	Tumour state: T cell (+)	✓	MIR155HG	
mir-335	0.36 (0.07)	0.36	0.58	0.14	0.68	Host gene	b	MEST	
mir-4728	0.58 (0.05)	0.59	0.50	0.31	0.67	Host gene + tumour state	✓	ERBB2	
mir-146a	0.64 (0.05)	0.67	0.46	0.54	0.64	Tumour state: T cell (+)	b	MIR3142HG	
mir-210	0.62 (0.09)	0.70	0.17	0.50	0.63	Host gene + tumour state	(✓)	MIR210HG	
mir-149	0.53 (0.03)	0.55	0.11	0.31	0.63	Own locus	(✓)	GPC1	
mir-3614	0.62 (0.08)	0.63	0.00	0.44	0.62	Own locus	b	TRIM25	
let-7c	0.60 (0.04)	0.62	0.50	0.52	0.62	Host gene	b	MIR99AHG	
mir-140	0.62 (0.04)	0.64	0.08	0.41	0.62	Host gene + tumour state	b	WWP2	
mir-92a-1	0.61 (0.10)	0.66	0.23	0.50	0.61	Tumour state: proliferation (+)	b	MIR17HG	P
mir-92a-2	0.61 (0.10)	0.66	–	0.50	0.61	Tumour state: proliferation (+)	b	AC002407.1	P H
mir-4772	0.60 (0.05)	0.62	0.50	0.56	0.61	Host gene	b	IL18RAP	
mir-378a	0.61 (0.08)	0.65	0.32	0.46	0.61	Host gene + tumour state	(✓)	PPARGC1B	
mir-598	0.28 (0.07)	0.32	0.51	0.15	0.60	Host gene	b	XKR6	
mir-106b	0.60 (0.10)	0.66	0.32	0.53	0.60	Tumour state: proliferation (+)	✓	MCM7	
mir-615	0.51 (0.04)	0.53	0.26	0.29	0.60	Host gene + tumour state	✓	AC012531.3;HOXC4;AC01…	
mir-342	0.51 (0.06)	0.56	0.35	0.29	0.60	Host gene + tumour state	✓	EVL	
mir-375	0.60 (0.07)	0.64	–	0.35	0.60	Own locus	b		
mir-139	0.58 (0.04)	0.59	0.48	0.48	0.59	Host gene	b	PDE2A	
mir-532	0.56 (0.06)	0.59	0.37	0.37	0.59	Host gene + tumour state	✓	CLCN5	
mir-130b	0.58 (0.07)	0.63	0.04	0.49	0.58	Tumour state: proliferation (+)	b	AP000553.1	
mir-1301	0.58 (0.06)	0.59	0.10	0.47	0.58	Tumour state: proliferation (+)	b	DNMT3A	
mir-30d	0.58 (0.07)	0.61	–	0.30	0.58	Own locus	b		
mir-1307	0.58 (0.11)	0.66	0.14	0.38	0.58	Host gene + tumour state	b	ATP5MD	
mir-10b	0.39 (0.06)	0.44	0.31	0.31	0.58	Host gene + tumour state	✓	HOXD3	
mir-3200	0.58 (0.03)	0.60	0.07	0.41	0.58	Residual signal	b	OSBP2	
mir-20a	0.58 (0.10)	0.68	0.19	0.49	0.58	Tumour state: ER/luminal (−)	b	MIR17HG	
mir-203a	0.58 (0.08)	0.60	–	0.22	0.58	Residual signal	b		
mir-223	0.57 (0.05)	0.60	0.24	0.44	0.57	Host gene + tumour state	b	MIR223HG	
mir-183	0.57 (0.05)	0.60	–	0.30	0.57	Own locus	b		
mir-142	0.57 (0.09)	0.64	0.06	0.50	0.57	Tumour state: T cell (+)	b	AC004687.1	
mir-148a	0.53 (0.07)	0.56	–	0.37	0.57	Own locus	✓		
mir-200c	0.56 (0.06)	0.62	0.10	0.25	0.56	Own locus	b	MIR200CHG	
mir-455	0.48 (0.08)	0.56	0.18	0.38	0.56	Host gene + tumour state	✓	COL27A1	
mir-542	0.43 (0.07)	0.46	–	0.27	0.56	Own locus	✓		
mir-625	0.48 (0.06)	0.52	0.20	0.23	0.56	Host gene + tumour state	✓	FUT8	
mir-937	0.55 (0.06)	0.58	0.11	0.39	0.55	Host gene + tumour state	(✓)	SCRIB	

miRNA	Full (SD)	Patient	Host	State	Ceiling	Category	Robust	Host gene(s)	Flags
mir-31	0.44 (0.07)	0.45	0.45	0.25	0.55	Host gene	b	MIR31HG	
mir-9-1	0.55 (0.07)	0.56	0.24	0.44	0.55	Host gene + tumour state	(✓)	MIR9-1HG	P
mir-9-2	0.55 (0.07)	0.55	–	0.44	0.55	Paralog locus	(✓)	LINC00461	P H
mir-9-3	0.55 (0.07)	0.56	0.02	0.44	0.55	Tumour state: ER/luminal (−)	b	MIR9-3HG	P
mir-584	0.55 (0.07)	0.57	–	0.42	0.55	Own locus	b	SH3TC2	H
mir-3150b	0.42 (0.06)	0.43	0.43	0.31	0.55	Host gene + tumour state	✓	MIR3150BHG;AC068189.1	
mir-942	0.51 (0.10)	0.54	0.15	0.46	0.54	Tumour state: ER/luminal (−)	b	TTF2	
mir-505	0.54 (0.07)	0.57	0.10	0.46	0.54	Tumour state: ER/luminal (−)	b	ATP11C	
mir-887	0.31 (0.08)	0.33	0.27	0.20	0.54	Host gene + tumour state	✓	FBXL7	
mir-141	0.53 (0.06)	0.60	0.08	0.27	0.53	Own locus	(✓)	MIR200CHG	
mir-99a	0.49 (0.06)	0.56	0.44	0.44	0.53	Host gene	b	MIR99AHG	
mir-1247	0.34 (0.07)	0.36	0.29	0.21	0.53	Host gene + tumour state	✓	DIO3OS	
mir-1270	0.21 (0.08)	0.24	0.32	0.12	0.53	Host gene + tumour state	✓	ZNF826P	
mir-181d	0.52 (0.02)	0.52	0.44	0.29	0.53	Host gene	b	NANOS3	
mir-200a	0.42 (0.09)	0.45	–	0.20	0.53	Own locus	✓		
mir-338	0.35 (0.09)	0.42	0.33	0.21	0.53	Host gene + tumour state	✓	AATK	
mir-193b	0.47 (0.06)	0.50	0.29	0.27	0.52	Host gene + tumour state	✓	MIR193BHG	
mir-29c	0.52 (0.08)	0.57	0.13	0.46	0.52	Tumour state: proliferation (−)	(✓)	MIR29B2CHG	X
mir-324	0.52 (0.04)	0.58	–	0.36	0.52	Own locus	b		
mir-221	0.50 (0.07)	0.52	0.36	0.30	0.52	Host gene + tumour state	✓	MIR222HG	
mir-3677	0.51 (0.15)	0.59	−0.07	0.39	0.51	Own locus	b	AC009065.4	
mir-301b	0.51 (0.06)	0.55	0.03	0.44	0.51	Tumour state: proliferation (+)	b	AP000553.1	
mir-196b	0.34 (0.13)	0.42	0.22	0.21	0.51	Host gene + tumour state	b	HOXA9;AC004080.3	
mir-301a	0.51 (0.09)	0.58	0.17	0.39	0.51	Host gene + tumour state	b	SKA2	
mir-345	0.51 (0.09)	0.57	–	0.37	0.51	Own locus	b		
mir-502	0.44 (0.07)	0.46	0.24	0.37	0.51	Host gene + tumour state	✓	CLCN5	X
mir-22	0.51 (0.05)	0.56	0.15	0.34	0.51	Own locus	b	MIR22HG	
mir-25	0.51 (0.11)	0.54	0.35	0.39	0.51	Host gene + tumour state	(✓)	MCM7	
mir-197	0.51 (0.07)	0.58	0.01	0.40	0.51	Host gene + tumour state	b	GNAI3	
mir-653	0.26 (0.08)	0.30	0.43	0.08	0.51	Host gene	b	CALCR	
mir-200b	0.41 (0.10)	0.46	–	0.19	0.51	Own locus	✓		
mir-758	0.50 (0.08)	0.54	–	0.39	0.50	Own locus	✓	MEG8	H
mir-93	0.50 (0.09)	0.60	0.28	0.41	0.50	Tumour state: proliferation (+)	b	MCM7	
mir-181c	0.47 (0.06)	0.52	0.40	0.26	0.50	Host gene + tumour state	s	NANOS3	
mir-30e	0.50 (0.07)	0.53	0.13	0.38	0.50	Host gene + tumour state	(✓)	NFYC	X
mir-127	0.48 (0.08)	0.53	–	0.36	0.50	Own locus	✓	AL117190.1	H
mir-4662a	0.41 (0.06)	0.44	–	0.11	0.50	Own locus	✓		
mir-877	0.50 (0.07)	0.51	0.07	0.41	0.50	Tumour state: proliferation (+)	b	ABCF1	
mir-501	0.49 (0.04)	0.53	0.27	0.36	0.49	Host gene + tumour state	✓	CLCN5	
mir-187	0.46 (0.04)	0.47	–	0.22	0.49	Own locus	✓		
mir-4677	0.49 (0.07)	0.50	−0.01	0.35	0.49	Host gene + tumour state	b	SDCCAG8	
mir-592	0.37 (0.06)	0.38	0.33	0.24	0.49	Host gene + tumour state	✓	GRM8	
mir-222	0.47 (0.10)	0.50	0.22	0.30	0.49	Host gene + tumour state	✓	MIR222HG	
mir-3127	0.49 (0.04)	0.50	0.13	0.27	0.49	Host gene + tumour state	b	CNNM4	
mir-21	0.49 (0.07)	0.55	0.15	0.34	0.49	Host gene + tumour state	b	VMP1	
mir-379	0.48 (0.11)	0.56	–	0.39	0.48	Tumour state: fibroblast (+)	b	MEG8	H
mir-182	0.48 (0.07)	0.54	–	0.25	0.48	Own locus	b		
mir-4661	0.45 (0.08)	0.46	0.21	0.30	0.48	Host gene + tumour state	✓	LRRC69	
mir-28	0.48 (0.04)	0.51	0.10	0.24	0.48	Host gene + tumour state	✓	LPP	
mir-1306	0.48 (0.07)	0.51	−0.02	0.42	0.48	Tumour state: proliferation (+)	(✓)	DGCR8	
mir-590	0.48 (0.11)	0.56	−0.05	0.44	0.48	Tumour state: ER/luminal (−)	✓	EIF4H	
mir-100	0.43 (0.07)	0.51	0.40	0.36	0.48	Host gene	b	MIR100HG	
mir-382	0.44 (0.07)	0.48	–	0.36	0.48	Own locus	✓		
mir-134	0.48 (0.08)	0.52	–	0.36	0.48	Own locus	b		
mir-199a-1	0.48 (0.08)	0.55	–	0.36	0.48	Paralog locus	b		X P
let-7d	0.48 (0.08)	0.47	−0.00	0.36	0.48	Own locus	b	MIRLET7A1HG	
let-7b	0.48 (0.07)	0.54	0.06	0.24	0.48	Residual signal	b	MIRLET7BHG	
mir-203b	0.48 (0.06)	0.51	–	0.19	0.48	Residual signal	b		
mir-381	0.44 (0.10)	0.49	–	0.37	0.47	Own locus	✓	MIR381HG	H
mir-431	0.42 (0.06)	0.44	–	0.32	0.47	Own locus	✓	AL117190.1	H

miRNA	Full (SD)	Patient	Host	State	Ceiling	Category	Robust	Host gene(s)	Flags
mir-23a	0.47 (0.07)	0.54	−0.03	0.33	0.47	Partly explained	b	AC020916.1	X
mir-146b	0.47 (0.11)	0.57	–	0.36	0.47	Residual signal	b		
mir-652	0.38 (0.06)	0.42	0.23	0.11	0.47	Host gene + tumour state	✓	TMEM164	
mir-181b-1	0.47 (0.04)	0.49	–	0.21	0.47	Residual signal	✓	MIR181A1HG	P H
mir-500a	0.44 (0.08)	0.50	0.26	0.32	0.47	Host gene + tumour state	✓	CLCN5	X
mir-199a-2	0.46 (0.08)	0.54	0.28	0.35	0.46	Host gene + tumour state	b	DNM3OS	X P
mir-95	0.31 (0.07)	0.34	0.27	0.06	0.46	Host gene + tumour state	✓	ABLIM2	
mir-153-2	0.37 (0.08)	0.38	0.36	0.26	0.46	Host gene + tumour state	(✓)	PTPRN2	P
mir-454	0.41 (0.08)	0.48	0.24	0.35	0.46	Host gene + tumour state	(✓)	SKA2	
mir-181b-2	0.46 (0.05)	0.48	0.07	0.22	0.46	Residual signal	✓	MIR181A2HG	P
mir-671	0.46 (0.05)	0.47	−0.02	0.34	0.46	Own locus	b	CHPF2	
mir-16-2	0.45 (0.14)	0.54	0.06	0.27	0.45	Residual signal	b	SMC4	P
mir-186	0.45 (0.13)	0.54	−0.04	0.36	0.45	Host gene + tumour state	b	ZRANB2	
mir-3074	0.45 (0.06)	0.50	–	0.28	0.45	Own locus	b		
mir-16-1	0.45 (0.14)	0.53	−0.07	0.27	0.45	Residual signal	b	DLEU2	P
mir-660	0.45 (0.10)	0.54	0.08	0.31	0.45	Host gene + tumour state	b	CLCN5	
mir-1266	0.45 (0.05)	0.50	0.07	0.21	0.45	Residual signal	b	MYO5C	
mir-92b	0.45 (0.09)	0.53	0.06	0.26	0.45	Own locus	b	THBS3-AS1	
mir-675	0.20 (0.05)	0.26	0.26	0.10	0.45	Host gene + tumour state	✓	H19	
mir-23b	0.43 (0.07)	0.50	0.23	0.22	0.45	Host gene + tumour state	✓	AOPEP	X
mir-370	0.36 (0.09)	0.42	–	0.29	0.45	Own locus	✓	AL117190.1;MEG8	H
mir-484	0.44 (0.09)	0.51	−0.04	0.29	0.44	Residual signal	b	NDE1	
mir-184	0.44 (0.06)	0.46	–	0.24	0.44	Residual signal	b		
mir-511	0.39 (0.09)	0.42	0.32	0.29	0.44	Host gene + tumour state	✓	MRC1	
mir-337	0.42 (0.06)	0.46	–	0.36	0.44	Tumour state: fibroblast (+)	b	AL117190.1	H
mir-29a	0.44 (0.08)	0.47	–	0.36	0.44	Tumour state: epithelial (−)	b	AC016831.6;AC016831.1	H
mir-199b	0.43 (0.09)	0.53	–	0.34	0.43	Partly explained	b		X
mir-27b	0.37 (0.08)	0.44	0.33	0.16	0.43	Host gene + tumour state	✓	AOPEP	X
mir-7-1	0.43 (0.06)	0.48	0.01	0.31	0.43	Host gene + tumour state	b	HNRNPK	P
mir-185	0.43 (0.07)	0.51	−0.00	0.26	0.43	Host gene + tumour state	b	TANGO2	
mir-2355	0.37 (0.07)	0.41	0.11	0.24	0.43	Host gene + tumour state	✓	KLF7	
let-7i	0.43 (0.07)	0.49	−0.02	0.34	0.43	Host gene + tumour state	b	AC048341.1	
mir-423	0.39 (0.05)	0.43	0.06	0.24	0.43	Host gene + tumour state	✓	NSRP1	
mir-151a	0.42 (0.14)	0.55	0.18	0.23	0.43	Host gene + tumour state	✓	PTK2	
mir-409	0.42 (0.04)	0.44	–	0.32	0.42	Own locus	✓		
mir-30b	0.42 (0.05)	0.47	–	0.24	0.42	Own locus	b		X
mir-96	0.42 (0.09)	0.49	–	0.26	0.42	Own locus	(✓)		
mir-493	0.38 (0.05)	0.41	–	0.31	0.42	Own locus	✓	AL117190.1	H
mir-15b	0.42 (0.08)	0.47	0.05	0.31	0.42	Host gene + tumour state	b	SMC4	
mir-194-2	0.42 (0.08)	0.46	–	0.07	0.42	Residual signal	✓	MIR194-2HG	P H
mir-181a-1	0.42 (0.04)	0.45	–	0.17	0.42	Residual signal	✓	MIR181A1HG	P H
mir-24-2	0.42 (0.07)	0.50	−0.01	0.38	0.42	Tumour state: ER/luminal (−)	(✓)	AC020916.1	P
mir-1468	0.42 (0.07)	0.43	0.10	0.28	0.42	Partly explained	b	ARHGEF9	
mir-744	0.37 (0.12)	0.45	0.07	0.19	0.41	Host gene + tumour state	✓	MAP2K4	
mir-421	0.41 (0.04)	0.44	0.06	0.34	0.41	Tumour state: ER/luminal (−)	b	FTX	
mir-194-1	0.41 (0.07)	0.44	–	0.06	0.41	Residual signal	✓		P
mir-24-1	0.41 (0.08)	0.49	−0.00	0.38	0.41	Tumour state: ER/luminal (−)	✓	AOPEP	P
mir-130a	0.41 (0.12)	0.51	0.21	0.24	0.41	Host gene + tumour state	b	AP000662.1	
mir-181a-2	0.41 (0.08)	0.45	0.22	0.20	0.41	Own locus	b	MIR181A2HG	P
mir-214	0.41 (0.05)	0.44	0.29	0.30	0.41	Host gene + tumour state	b	DNM3OS	
mir-769	0.40 (0.09)	0.47	−0.02	0.24	0.40	Residual signal	b	CCDC61	
mir-429	0.38 (0.09)	0.44	–	0.18	0.40	Own locus	✓		
mir-766	0.37 (0.16)	0.45	0.11	0.27	0.40	Host gene + tumour state	✓	SEPTIN6	
mir-3065	0.34 (0.05)	0.34	–	0.12	0.40	Own locus	✓		
mir-483	0.15 (0.07)	0.21	0.36	0.13	0.40	Host gene	b	IGF2;AC132217.2;INS-I…	
mir-582	0.40 (0.04)	0.42	0.23	0.17	0.40	Host gene + tumour state	b	PDE4D	
mir-125b-2	0.39 (0.10)	0.48	0.27	0.31	0.40	Host gene + tumour state	b	MIR99AHG	P
mir-4746	0.40 (0.06)	0.41	–	0.30	0.40	Own locus	(✓)		
mir-204	0.40 (0.05)	0.43	–	0.34	0.40	Tumour state: basal (+)	b	TRPM3	H
mir-576	0.40 (0.09)	0.46	−0.06	0.30	0.40	Host gene + tumour state	b	SEC24B	

miRNA	Full (SD)	Patient	Host	State	Ceiling	Category	Robust	Host gene(s)	Flags
mir-369	0.37 (0.07)	0.40	–	0.31	0.40	Own locus	s		
mir-363	0.40 (0.06)	0.41	–	0.13	0.40	Own locus	b	AC002407.1	H
mir-589	0.40 (0.08)	0.43	0.01	0.22	0.40	Own locus	b	FBXL18	
mir-708	0.40 (0.08)	0.45	0.23	0.23	0.40	Host gene + tumour state	✓	TENM4	
mir-940	0.40 (0.12)	0.45	−0.04	0.30	0.40	Partly explained	b	AC009065.4	
mir-500b	0.35 (0.09)	0.39	0.16	0.29	0.39	Host gene + tumour state	✓	CLCN5	X
mir-188	0.36 (0.11)	0.42	0.19	0.31	0.39	Tumour state: proliferation (+)	b	CLCN5	
mir-33a	0.38 (0.11)	0.45	0.03	0.31	0.39	Host gene + tumour state	✓	SREBF2	
mir-99b	0.39 (0.10)	0.44	−0.06	0.26	0.39	Host gene + tumour state	b	SPACA6	
mir-98	0.39 (0.05)	0.40	−0.01	0.25	0.39	Residual signal	b	HUWE1	
mir-125b-1	0.38 (0.10)	0.48	0.32	0.30	0.39	Host gene	b	MIR100HG	P
mir-218-2	0.39 (0.07)	0.41	0.07	0.27	0.39	Paralog locus	b	SLIT3	P
mir-1245a	0.33 (0.06)	0.36	0.33	0.30	0.39	Host gene	b	COL3A1	
mir-1287	0.39 (0.10)	0.48	0.05	0.14	0.39	Host gene + tumour state	✓	PYROXD2	
mir-889	0.38 (0.09)	0.41	–	0.31	0.38	Own locus	s	MIR381HG	H
mir-33b	0.32 (0.16)	0.42	0.07	0.25	0.38	Host gene + tumour state	✓	SREBF1	
mir-20b	0.38 (0.06)	0.44	–	0.15	0.38	Own locus	✓	AC002407.1	H
mir-218-1	0.38 (0.07)	0.40	0.15	0.27	0.38	Host gene + tumour state	b	SLIT2	P
mir-410	0.36 (0.12)	0.42	–	0.29	0.38	Own locus	✓		
mir-192	0.38 (0.10)	0.46	–	0.18	0.38	Residual signal	b	MIR194-2HG	X H
mir-432	0.31 (0.09)	0.37	–	0.25	0.38	Own locus	✓	AL117190.1	H
mir-487b	0.34 (0.08)	0.36	–	0.27	0.38	Own locus	✓	MIR381HG	H
mir-107	0.37 (0.09)	0.42	−0.01	0.22	0.37	Residual signal	b	PANK1	X
mir-3170	0.37 (0.06)	0.38	0.06	0.27	0.37	Host gene + tumour state	(✓)	FARP1	
mir-874	0.33 (0.08)	0.41	0.06	0.19	0.37	Host gene + tumour state	✓	KLHL3	
mir-629	0.37 (0.07)	0.40	0.03	0.17	0.37	Host gene + tumour state	b	TLE3	
mir-425	0.37 (0.07)	0.43	−0.07	0.19	0.37	Residual signal	b	DALRD3	
mir-550a-1	0.32 (0.08)	0.33	−0.01	0.22	0.37	Paralog locus	✓	ZNRF2	P
mir-361	0.37 (0.11)	0.44	−0.05	0.23	0.37	Host gene + tumour state	b	CHM	
mir-642a	0.33 (0.06)	0.37	0.07	0.17	0.37	Host gene + tumour state	✓	GIPR	
mir-19b-1	0.36 (0.20)	0.54	0.10	0.32	0.36	Tumour state: fibroblast (−)	b	MIR17HG	P
mir-5683	0.36 (0.06)	0.37	–	0.28	0.36	Partly explained	b		
mir-1180	0.30 (0.13)	0.43	0.19	0.16	0.36	Host gene + tumour state	✓	B9D1	
mir-485	0.28 (0.06)	0.30	–	0.23	0.36	Own locus	✓		
mir-191	0.36 (0.09)	0.43	−0.07	0.21	0.36	Partly explained	b	DALRD3	
mir-5586	0.36 (0.07)	0.37	0.10	0.24	0.36	Host gene + tumour state	b	RTN1	
mir-424	0.26 (0.09)	0.26	0.09	0.16	0.36	Host gene + tumour state	✓	MIR503HG	
mir-29b-2	0.36 (0.10)	0.46	−0.04	0.30	0.36	Tumour state: fibroblast (−)	b	MIR29B2CHG	P
mir-616	0.36 (0.07)	0.41	−0.03	0.31	0.36	Tumour state: proliferation (+)	b	DDIT3;AC022506.1	
mir-760	0.36 (0.16)	0.44	–	0.17	0.36	Residual signal	b		
mir-362	0.35 (0.13)	0.45	0.07	0.27	0.35	Host gene + tumour state	b	CLCN5	
mir-3934	0.32 (0.07)	0.35	–	0.27	0.35	Own locus	(✓)		
mir-154	0.29 (0.06)	0.32	–	0.25	0.35	Own locus	✓		
mir-328	0.35 (0.11)	0.45	–	0.23	0.35	Own locus	b		
mir-550a-2	0.30 (0.07)	0.31	0.01	0.21	0.35	Host gene + tumour state	b	ZNRF2P1	P
mir-195	0.35 (0.08)	0.42	–	0.21	0.35	Own locus	b	AC040977.2;MIR497HG	H
mir-19a	0.35 (0.20)	0.53	0.02	0.33	0.35	Tumour state: proliferation (+)	b	MIR17HG	
mir-411	0.30 (0.12)	0.36	–	0.26	0.35	Own locus	✓	MEG8	H
mir-132	0.35 (0.07)	0.40	–	0.17	0.35	Residual signal	b		
mir-29b-1	0.34 (0.10)	0.43	–	0.29	0.34	Tumour state: fibroblast (−)	b	AC016831.6;AC016831.1	X P H
mir-377	0.26 (0.08)	0.29	–	0.22	0.34	Own locus	✓		
mir-1296	0.33 (0.05)	0.37	−0.03	0.21	0.33	Partly explained	b	JMJD1C	
mir-106a	0.33 (0.11)	0.42	–	0.20	0.33	Own locus	b	AC002407.1	X H
mir-654	0.32 (0.07)	0.34	–	0.26	0.33	Own locus	s		
mir-128-1	0.33 (0.07)	0.39	0.05	0.22	0.33	Host gene + tumour state	b	R3HDM1	P
mir-30c-2	0.29 (0.12)	0.45	0.16	0.10	0.32	Host gene + tumour state	✓	AL136164.2;LINC00472	X P
mir-296	−0.00 (0.07)	0.07	0.20	−0.02	0.32	Host gene + tumour state	✓	AL132655.2	
mir-19b-2	0.32 (0.21)	0.50	–	0.28	0.32	Tumour state: fibroblast (−)	b	AC002407.1	P H
mir-299	0.31 (0.06)	0.30	–	0.26	0.32	Tumour state: fibroblast (+)	b	MEG8	H
mir-378c	0.31 (0.12)	0.40	–	0.26	0.31	Tumour state: ER/luminal (−)	b		

miRNA	Full (SD)	Patient	Host	State	Ceiling	Category	Robust	Host gene(s)	Flags
mir-103a-2	0.31 (0.15)	0.47	−0.12	0.14	0.31	Residual signal	b	PANK2	X P
mir-103a-1	0.31 (0.15)	0.47	−0.12	0.14	0.31	Residual signal	b	PANK3	X P
mir-148b	0.31 (0.12)	0.48	−0.09	0.24	0.31	Host gene + tumour state	b	AC078778.1;COPZ1	
mir-340	0.31 (0.12)	0.40	0.02	0.17	0.31	Host gene + tumour state	b	RNF130	
mir-495	0.25 (0.06)	0.29	–	0.20	0.31	Own locus	s	MEG8	H
mir-3928	0.31 (0.10)	0.33	–	0.25	0.31	Tumour state: proliferation (+)	b		
mir-450b	0.20 (0.07)	0.29	–	0.10	0.31	Own locus	(✓)		
mir-374a	0.31 (0.07)	0.38	−0.05	0.18	0.31	Residual signal	b	FTX	
mir-628	0.31 (0.14)	0.45	−0.07	0.24	0.31	Own locus	b	CCPG1;DNAAF4-CCPG1	
mir-539	0.30 (0.10)	0.36	–	0.23	0.30	Own locus	b	MIR381HG	H
mir-152	0.23 (0.14)	0.40	0.20	0.14	0.30	Poorly predicted	b	COPZ2	
mir-145	0.30 (0.10)	0.37	0.10	0.14	0.30	Poorly predicted	b	CARMN	
mir-15a	0.30 (0.15)	0.46	−0.01	0.11	0.30	Poorly predicted	b	DLEU2	
mir-136	0.27 (0.09)	0.37	–	0.16	0.29	Poorly predicted	b	AL117190.1	H
mir-496	0.29 (0.08)	0.32	–	0.23	0.29	Poorly predicted	b		
mir-330	0.29 (0.16)	0.46	−0.10	0.04	0.29	Poorly predicted	b	EML2	
mir-193a	0.29 (0.10)	0.37	–	0.06	0.29	Poorly predicted	b		
mir-4668	0.24 (0.09)	0.31	0.09	0.18	0.29	Poorly predicted	b	UGCG	
mir-574	0.28 (0.08)	0.33	0.04	0.16	0.28	Poorly predicted	b	FAM114A1	
mir-365a	0.28 (0.05)	0.34	0.11	0.16	0.28	Poorly predicted	b	MIR193BHG	X
mir-323a	0.23 (0.12)	0.28	–	0.16	0.28	Poorly predicted	b	MEG8	H
mir-1271	0.22 (0.12)	0.30	0.11	0.19	0.28	Poorly predicted	b	ARL10	
mir-376c	0.28 (0.05)	0.31	–	0.18	0.28	Poorly predicted	b		X
mir-365b	0.28 (0.05)	0.34	–	0.16	0.28	Poorly predicted	b	AC003101.1	X H
mir-7706	0.27 (0.07)	0.28	−0.02	0.18	0.27	Poorly predicted	b	AKAP13	
mir-4638	0.27 (0.11)	0.32	–	0.21	0.27	Poorly predicted	b		
let-7g	0.27 (0.12)	0.37	−0.07	0.13	0.27	Poorly predicted	b	WDR82	
mir-125a	0.27 (0.11)	0.37	−0.05	0.10	0.27	Poorly predicted	b	SPACA6	
mir-651	0.22 (0.10)	0.26	−0.01	0.14	0.27	Poorly predicted	b	AC079142.1	
mir-487a	0.19 (0.04)	0.20	–	0.19	0.27	Poorly predicted	b		
mir-3613	0.27 (0.11)	0.34	−0.02	0.16	0.27	Poorly predicted	b	DLEU2	
mir-27a	0.26 (0.17)	0.39	0.03	0.22	0.26	Poorly predicted	b	AC020916.1	X
mir-219a-1	0.26 (0.05)	0.29	–	0.14	0.26	Poorly predicted	b		P
mir-655	0.21 (0.05)	0.24	–	0.18	0.26	Poorly predicted	b	MIR381HG	H
mir-101-2	0.26 (0.14)	0.43	−0.07	0.21	0.26	Poorly predicted	b	RCL1	P
mir-503	0.16 (0.11)	0.25	0.03	0.05	0.26	Poorly predicted	b	MIR503HG	
mir-34a	0.21 (0.11)	0.35	−0.00	0.03	0.26	Poorly predicted	b	MIR34AHG	
mir-128-2	0.23 (0.11)	0.35	–	0.16	0.26	Poorly predicted	b	ARPP21	P H
mir-450a-1	0.14 (0.07)	0.17	–	0.09	0.25	Poorly predicted	b		P
mir-101-1	0.25 (0.14)	0.42	−0.10	0.21	0.25	Poorly predicted	b	JAK1	P
mir-320a	0.25 (0.07)	0.34	–	0.05	0.25	Poorly predicted	b		X
mir-34c	0.24 (0.08)	0.27	–	0.12	0.24	Poorly predicted	b	AP002008.4	H
mir-497	0.24 (0.09)	0.31	–	0.08	0.24	Poorly predicted	b	AC040977.2;MIR497HG	H
mir-374b	0.24 (0.06)	0.32	−0.06	0.11	0.24	Poorly predicted	b	FTX	
mir-450a-2	0.12 (0.05)	0.18	–	0.08	0.24	Poorly predicted	b		P
mir-320b-2	0.24 (0.02)	0.25	−0.01	0.13	0.24	Poorly predicted	✓	NVL	X P
mir-26a-1	0.23 (0.17)	0.34	−0.04	0.15	0.23	Poorly predicted	b	CTDSPL	P
mir-126	0.23 (0.11)	0.29	−0.04	0.14	0.23	Poorly predicted	b	EGFL7	
mir-339	0.23 (0.09)	0.33	−0.08	0.12	0.23	Poorly predicted	b	C7orf50	
mir-26a-2	0.23 (0.17)	0.34	−0.03	0.15	0.23	Poorly predicted	b	CTDSP2	P
mir-664a	0.23 (0.11)	0.37	−0.09	0.12	0.23	Poorly predicted	b	RAB3GAP2;SNORA36B	
mir-1-2	0.23 (0.08)	0.28	–	0.09	0.23	Poorly predicted	b	MIR133A1HG	P H
mir-323b	0.19 (0.16)	0.28	–	0.18	0.23	Poorly predicted	b		
let-7a-3	0.23 (0.15)	0.39	−0.01	0.09	0.23	Poorly predicted	b	MIRLET7BHG	P
let-7a-1	0.23 (0.15)	0.39	−0.10	0.09	0.23	Poorly predicted	b	MIRLET7A1HG	P
mir-3607	0.23 (0.11)	0.34	–	0.16	0.23	Poorly predicted	b		
let-7a-2	0.23 (0.16)	0.39	−0.11	0.09	0.23	Poorly predicted	b	MIR100HG	P
mir-1976	0.22 (0.10)	0.37	0.11	0.13	0.22	Poorly predicted	b	RPS6KA1	
mir-32	0.22 (0.12)	0.31	−0.04	0.17	0.22	Poorly predicted	b	TMEM245	
mir-331	0.19 (0.23)	0.39	–	0.17	0.22	Poorly predicted	b		

miRNA	Full (SD)	Patient	Host	State	Ceiling	Category	Robust	Host gene(s)	Flags
mir-451a	0.21 (0.05)	0.23	–	0.11	0.21	Poorly predicted	✓	AC024267.1	H
mir-190a	0.10 (0.11)	0.20	0.07	−0.01	0.21	Poorly predicted	b	TLN2	
mir-144	0.21 (0.05)	0.21	–	0.11	0.21	Poorly predicted	✓	AC024267.1	H
mir-548v	0.12 (0.04)	0.14	0.05	0.05	0.21	Poorly predicted	✓	MTUS1	
mir-486-1	0.21 (0.07)	0.30	0.09	0.11	0.21	Poorly predicted	✓	ANK1	P
mir-1277	0.19 (0.11)	0.25	−0.05	0.20	0.21	Poorly predicted	b	WDR44	
mir-217	0.20 (0.06)	0.24	–	0.08	0.20	Poorly predicted	✓	MIR217HG	H
mir-6511b-1	0.20 (0.05)	0.23	−0.03	0.12	0.20	Poorly predicted	✓	PKD1	P
mir-5000	0.20 (0.06)	0.22	–	0.11	0.20	Poorly predicted	✓		
mir-1248	0.19 (0.05)	0.25	−0.02	0.14	0.19	Poorly predicted	✓	EIF4A2;SNORA81	
mir-486-2	0.19 (0.06)	0.28	–	0.11	0.19	Poorly predicted	✓		P
mir-491	0.15 (0.13)	0.19	0.06	0.06	0.18	Poorly predicted	b	FOCAD	
mir-30c-1	0.18 (0.19)	0.40	−0.12	0.03	0.18	Poorly predicted	b	NFYC	X P
mir-320b-1	0.18 (0.03)	0.19	–	0.10	0.18	Poorly predicted	✓		X P
mir-212	0.18 (0.10)	0.27	–	0.14	0.18	Poorly predicted	b		
mir-6892	0.18 (0.06)	0.23	−0.03	0.09	0.18	Poorly predicted	✓	ZYX	
mir-3913-2	0.08 (0.03)	0.09	–	0.04	0.18	Poorly predicted	✓		P
mir-6511b-2	0.18 (0.05)	0.20	−0.02	0.11	0.18	Poorly predicted	✓	PKD1P6	P
mir-3913-1	0.09 (0.05)	0.12	–	0.04	0.18	Poorly predicted	✓		P
mir-1-1	0.17 (0.07)	0.22	–	0.08	0.17	Poorly predicted	✓	MIR1-1HG	P H
mir-664b	0.15 (0.05)	0.20	−0.04	0.09	0.17	Poorly predicted	✓	DKC1;SNORA36A	
mir-508	0.17 (0.07)	0.19	–	0.03	0.17	Poorly predicted	✓		
mir-215	0.17 (0.16)	0.29	–	0.05	0.17	Poorly predicted	b		X
mir-3605	0.17 (0.13)	0.31	−0.09	0.15	0.17	Poorly predicted	b	PHC2	
let-7e	0.17 (0.09)	0.27	0.03	0.02	0.17	Poorly predicted	✓	SPACA6	
mir-376b	0.12 (0.06)	0.14	–	0.09	0.17	Poorly predicted	✓		X
mir-494	0.11 (0.05)	0.16	–	0.11	0.15	Poorly predicted	✓	MEG8	H
mir-412	0.11 (0.04)	0.12	–	0.10	0.15	Poorly predicted	✓		
mir-659	0.15 (0.05)	0.16	0.01	0.08	0.15	Poorly predicted	✓	ANKRD54	
mir-551b	0.14 (0.10)	0.17	–	0.06	0.14	Poorly predicted	✓	EGFEM1P	H
mir-3615	0.12 (0.18)	0.25	−0.07	0.08	0.13	Poorly predicted	b	SLC9A3R1	
mir-6842	0.01 (0.12)	0.17	0.04	−0.01	0.12	Poorly predicted	b	PTK2B	
mir-2110	0.11 (0.09)	0.23	−0.04	0.07	0.12	Poorly predicted	✓	CCDC186	
mir-3917	0.12 (0.07)	0.19	0.07	0.06	0.12	Poorly predicted	✓	STMN1	
mir-151b	0.11 (0.05)	0.14	–	0.04	0.11	Poorly predicted	✓		
mir-26b	0.11 (0.09)	0.19	−0.06	−0.00	0.11	Poorly predicted	✓	CTDSP1	
mir-4326	0.08 (0.07)	0.13	−0.02	0.07	0.10	Poorly predicted	✓	ARFGAP1	
mir-326	0.10 (0.19)	0.32	−0.07	0.00	0.10	Poorly predicted	b	ARRB1	
mir-1249	0.10 (0.14)	0.20	−0.07	0.06	0.10	Poorly predicted	✓	KIAA0930	
mir-34b	0.08 (0.09)	0.13	–	−0.02	0.08	Poorly predicted	✓	AP002008.4	H
mir-627	0.08 (0.07)	0.14	−0.03	0.02	0.08	Poorly predicted	✓	VPS39	
mir-3653	0.07 (0.11)	0.26	–	0.07	0.07	Poorly predicted	✓		
mir-143	0.01 (0.21)	0.37	−0.04	−0.01	0.01	Poorly predicted	b	CARMN	
mir-3130-2	−0.05 (0.15)	0.13	−0.08	−0.04	−0.04	Poorly predicted	✓	FASTKD2	P
let-7f-2	−0.05 (0.25)	0.28	−0.23	−0.15	−0.05	Poorly predicted	✓	HUWE1	P
let-7f-1	−0.06 (0.26)	0.28	−0.15	−0.15	−0.06	Poorly predicted	✓	MIRLET7A1HG	P
mir-3130-1	−0.10 (0.16)	0.09	–	−0.07	−0.07	Poorly predicted	✓		P

Supplementary Table S2. Four miRNAs absent from many tumours, below the expression threshold for the main analysis, evaluated with the same models (patient-grouped folds). All four were assigned to the host gene.

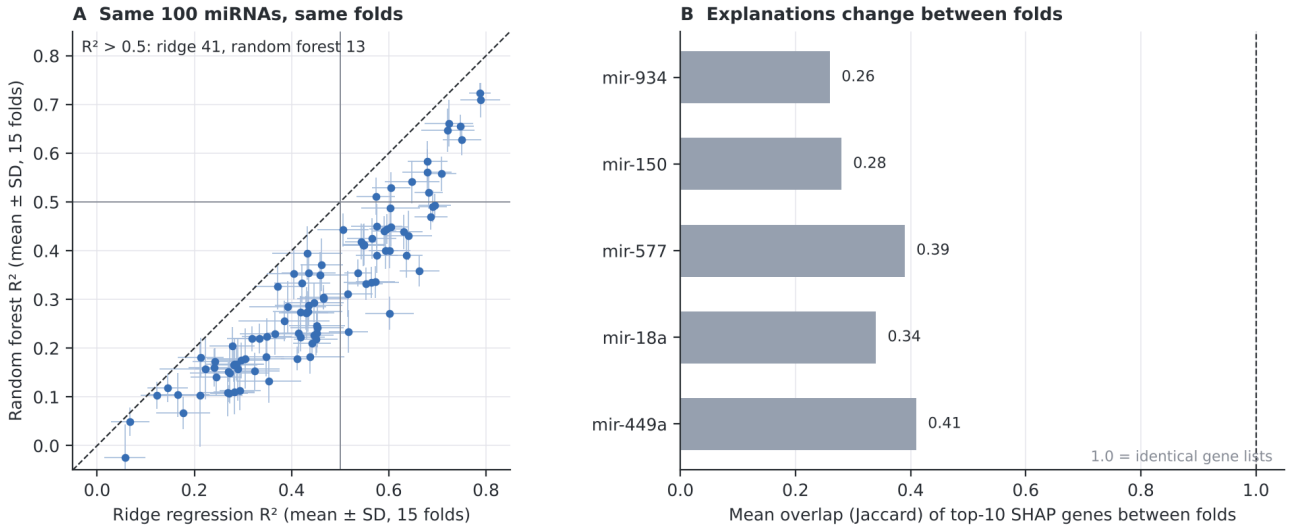
miRNA	Host gene	Tumours with RPM = 0	Full R^2	Host R^2	State R^2	Host + state R^2
mir-934	VGLL1	43%	0.79	0.89	0.70	0.91
mir-577	UGT8	23%	0.72	0.73	0.67	0.80
mir-449a	CDC20B	37%	0.69	0.85	0.29	0.87
mir-449b	CDC20B	59%	0.65	0.79	0.24	0.81

Supplementary Table S3. Marker-gene programmes used to interpret principal components and miRNAs.

Programme	Marker genes
proliferation	MKI67, TOP2A, CCNB1, BUB1, CDK1, MCM2, PCNA
T cell	CD3D, CD3E, CD2, CD247
B/plasma	MS4A1, CD19, CD79A, MZB1, JCHAIN
myeloid	CD68, CD163, CSF1R, CD14
fibroblast	COL1A1, COL1A2, DCN, LUM
endothelial	PECAM1, VWF, CDH5
ER/luminal	ESR1, PGR, FOXA1, GATA3
basal	KRT5, KRT14, KRT17
HER2	ERBB2, GRB7
hypoxia	CA9, VEGFA, ADM, NDRG1
epithelial	EPCAM, CDH1, KRT8, KRT18
ciliary	FOXJ1, CDHR3, DNAH5

Supplementary Note: model choice

Before the main analysis, two model classes were compared for the 100 most variable miRNA precursors in all 1,087 matched tumours (20,513 genes with TPM > 1 in at least 20% of samples), on the same patient-grouped folds. A random forest (Breiman, 2001; 100 trees, max_features = "sqrt", otherwise default settings) reached $R^2 > 0.5$ for 13 of the 100 miRNAs and ridge regression for 41, and ridge regression performed better for 100 of them (Supplementary Figure S1A). This comparison applies to the random-forest configuration used and not to random forests in general. Gene-level explanations of the random forest were unstable: SHAP values (TreeExplainer; Lundberg et al., 2020) were computed on each test fold of one repeat, and the ten genes with the highest mean |SHAP| overlapped by only 0.26–0.41 (mean pairwise Jaccard index between folds) for the five best-predicted miRNAs (Supplementary Figure S1B). The only consistently top-ranked genes were host genes of intronic miRNAs (for example VGLL1 for mir-934, ranked 2–12 across folds). Ridge regression was therefore used for all further analyses, and predictions were interpreted by comparison with reference models rather than by ranking individual genes. This comparison used patient-grouped folds and a different cohort and gene filter from the main analysis.



Supplementary Figure S1. Model comparison for the 100 most variable miRNA precursors (1,087 tumours, 20,513 genes, patient-grouped folds). (A) Random forest (default settings) versus ridge regression; points, mean R^2 over 15 test folds; bars, SD; dashed line, equal performance. (B) Stability of SHAP explanations: mean pairwise Jaccard overlap of the top-10 genes by mean |SHAP| between the five folds of one cross-validation repeat.

Data and code availability

TCGA-BRCA expression data are openly available from the Genomic Data Commons (portal.gdc.cancer.gov); the manifest and sample sheet used here are in the repository. PAM50 calls and purity estimates are distributed with the TCGAbiolinks Bioconductor package, and GENCODE v36 is available from gencodegenes.org. All analysis code, the per-miRNA results tables and the figures are available at https://github.com/janaki-bio/mirna_mrna_research.

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